Contradiction Fields: CRT Rigidity, Prime Matrix Geometry, and Zeta-Zero Escape

Project manuscript

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Abstract

This monograph draft unifies two contradiction-field research programs. The first studies prime existence in CRT prime matrices through row, column, diagonal, anchor, and rough-composite rigidity. The second studies an off-critical zero of the Riemann zeta function through a global contradiction-field ledger for prime anomalies. The common method is to convert a hypothetical counterexample into a structured covering field, decompose it into local rigidity layers, and exclude all free escape channels by capacity, periodicity, and no-cycle bookkeeping.

This is a synthesis and dependency manuscript. It does not assert that the Riemann Hypothesis or the prime-matrix row/column theorem has already been accepted as a final unconditional theorem. Each claim is marked by status in the accompanying status table: proved in manuscript, reduced to a structured input, external, computationally certified, or requiring independent referee verification.

Contents

Chapter 1

Guiding Principle: The Contradiction Field

A contradiction field is a structured ledger attached to a hypothetical counterexample. Instead of estimating a single exceptional set, it records every possible explanation for the counterexample as a named channel. A proof closes when every channel is either absorbed, forced to descend in a finite potential, transferred with source deletion, or contradicts a global capacity lower/upper bound.

The common rigidities used throughout this manuscript are:

1. CRT skeletons and nonzero-residue balance;
2. local short-window non-reuse of large prime factors;
3. small-prime shell migration and diagonal locking;
4. rough-composite and double-anchor decompositions;
5. capacity ledgers after baseline subtraction;
6. no-cycle potentials for internal escape mechanisms.

Chapter 2

Prime-Matrix Contradiction Fields

Let P be an odd prime and write the $P \times P$ prime matrix entries as

$$n(r, c) = (r - 1)P + c, \qquad 1 \leq r, c \leq P.$$

The row and column programs attempt to show that every nontrivial row or column contains a prime. The current formal appendix reduces a row or non-P column counterexample to a structured bad configuration after removing small-factor locks and tail anchors.

Reduction-closed Statement 2.1 (A/B reduction, recorded form)**.** *A nontrivial full-composite row or column, after small-factor and tail-anchor peeling, induces a Structured-EHPD bad configuration satisfying CRT balance, large-factor non-reuse, and anchor-capacity constraints.*

Reference. See `docs/row-column-reduction-formal-appendix.md` and the theorem list

`docs/monograph/prime-matrix-ab-entrance-theorem-list.md`.

This statement is a reduction interface; it does not by itself prove the Structured-EHPD exclusion theorem. $\square$

2.1 Two-Point Rough-Pair Propositions

A natural next strengthening asks for two simultaneous nonzero-residue constraints in the back half of the prime matrix. Fix an even integer $w > 0$ and ask whether there is an entry x in the back half of the $P \times P$ matrix such that both x and $x - w$ are nonzero modulo every prime $p \leq P$, or equivalently both x and $w - x$ are nonzero modulo every prime $p \leq P$.

Proposition 2.2 (Equivalence and strength of the two-point rough-pair problem)**.** *For fixed even $w > 0$, the two conditions*

$$p \nmid x(x - w) \quad (p \leq P)$$

and

$$p \nmid x(w - x) \quad (p \leq P)$$

are identical. In the region $1 < x \leq P^2$, any integer nonzero modulo every prime $p \leq P$ is prime. Consequently, a proof of this two-point assertion for every sufficiently large P would imply infinitely many prime pairs of gap w; in particular $w = 2$ would imply the twin-prime conjecture.

Proof. For each $p \le P$, the condition $p \nmid x - w$ is $x \not\equiv w \pmod p$, and the condition $p \nmid w - x$ is the same congruence. Both formulations therefore exclude exactly the two residue classes 0 and w modulo each p, with possible collision when $p \mid w$.

If $1 < x \le P^2$ is nonzero modulo every prime $p \le P$ and is composite, then it has a prime factor $q \le \sqrt{x} \le P$, contradicting the nonzero-residue condition. Thus such an x is prime. If the two-point assertion holds for all sufficiently large $P > w$, then x and $x - w$ are both prime in infinitely many growing back-half matrix intervals, giving infinitely many prime pairs of gap w. □

Remark 2.3 (Status)**.** This two-point problem is therefore not a routine corollary of the one-point row/column contradiction field. It is a prime-pair-level strengthening, naturally expressible as a two-point rough Jacobsthal problem. In this monograph it is recorded as a future contradiction-field direction, not as an unconditional theorem.

2.2 Two-Point Secondary-Sieve Contradiction Field

The two-point rough-pair problem admits a more detailed contradiction-field research program. The current working manuscript is recorded in

`docs/monograph/two-point-secondary-sieve-research.md`.

It studies the secondary sieve condition

$$x \not\equiv 0, w \pmod p \qquad (p \le P)$$

inside back-half matrix windows, and tries to exclude a two-point rough-pair counterexample by a hierarchy of local rigidity fields.

Definition 2.4 (Two-point secondary-sieve field)**.** Fix an even integer $w > 0$. A two-point secondary-sieve field is the structured ledger obtained from a hypothetical interval in which every candidate x fails at least one of the two congruence conditions $p \nmid x$ or $p \nmid x - w$ for some prime $p \le P$. Its named channels are small-prime double locks, top-scale large-factor coverage, short-block zero-mass loss, multiplicative near-crossings, Fourier CRT defects, and smoothing boundary terms.

Lemma 2.5 (Fixed-order local crossing complexity)**.** *In the secondary-sieve manuscript, every fixed-order local crossing test is expanded only through truncated hard-skeleton weights*

$$d \le R, \qquad R \le (\log P)^{C_R},$$

with coefficient mass $\sum_{d \le R} |\lambda_d| \le (\log P)^{C_\lambda}$ *and short-block shifts* $|h| < L \asymp (\log\log P)^2$*. For every fixed order k, the effective historical complexity satisfies*

$$Q_{\mathrm{eff},k} \ll_k \left(\sum_{d \le R} |\lambda_d| \right)^k R^k L^{O(k)} \le (\log P)^{C_k}.$$

In particular the required $k = 2, 3$ local crossing tests have polylogarithmic effective complexity.

Proof. After expanding the k truncated weights, only moduli $d_1, \ldots, d_k \le R$ occur. For fixed new-line primes p_i, phases, and block shifts, all conditions are one-variable linear congruences. By the Chinese remainder theorem the local system is either inconsistent or occupies one residue class modulo a divisor of

$$\operatorname{lcm}(d_1, \ldots, d_k) \operatorname{lcm}(p_1, \ldots, p_k).$$

The primes p_i are the explicit variables being summed in the crossing estimate; they contribute the usual Y^k scale, not hidden historical complexity. The historical part is bounded by $\operatorname{lcm}(d_1, \ldots, d_k) \le R^k$, the block shifts contribute $L^{O(k)}$ patterns, and the coefficient expansion contributes $(\sum |\lambda_d|)^k$. This proves the displayed bound. The true residual set U_Y is not expanded into a full primorial period. □

Lemma 2.6 (Zero-mass and smoothing discharge)**.** *Assume the SC2/diagonal package supplies the moment bounds*

$$M_1 := \frac{\sum_B X_B R_B}{\sum_B X_B} \le 0.45, \qquad M_2 := \frac{\sum_B X_B \binom{R_B}{2}}{\sum_B X_B} \ge 0.05,$$

and

$$M_3 := \frac{\sum_B X_B \binom{R_B}{3}_+}{\sum_B X_B} \le 0.03.$$

Then the weighted zero-hit mass satisfies

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \ge 0.57.$$

Moreover, if the directional-balance endpoint estimate gives

$$N_{\text{endpoint}}/N_{\text{total}} \le C_{\text{end}}\Delta + o(1),$$

then choosing $\Delta \le 0.01/C_{\text{end}}$ gives a smooth-to-sharp loss < 0.03 for all sufficiently large P.

Proof. For every integer $R \ge 0$,

$$1_{R=0} \ge 1 - R + \binom{R}{2} - \binom{R}{3}_+.$$

Multiplying by $X_B \ge 0$ and summing over blocks gives

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \ge 1 - M_1 + M_2 - M_3 \ge 1 - 0.45 + 0.05 - 0.03 = 0.57.$$

For smoothing, choose even smooth cutoffs $W_- \le 1_{|t|<1} \le W_+$ whose difference is supported in $1 - \Delta \le |t| \le 1 + \Delta$. The sharp and smoothed counts differ only on this endpoint layer. The displayed directional-balance bound therefore gives loss at most $C_{\text{end}}\Delta + o(1)$, which is < 0.03 after the stated choice of Δ and then taking P large. □

Lemma 2.7 (Adaptive true-hit layering dichotomy)**.** *Let U_Y be the current true residual set and define, for each new prime $p > Y$,*

$$a_p = \frac{1}{|U_Y|} \#\{x \in U_Y : x \equiv 0 \text{ or } w \pmod p\}.$$

Fix $\eta = 0.05$. Either some p has $a_p > \eta$, in which case the configuration is a Single-Prime CRTDefect, or the remaining primes can be greedily partitioned into consecutive layers $\mathcal{P}_j$ satisfying

$$\nu_j^{\text{real}} := \sum_{p \in \mathcal{P}_j} a_p \le 0.4,$$

and, except for the final layer, $\nu_j^{\text{real}} > 0.4 - \eta \ge 0.35$.

Proof. If $a_p > \eta$ for some p, then a fixed positive proportion of U_Y lies in the two residue classes $0, w$ modulo p. This is exactly the Single-Prime CRTDefect exit, to be excluded by the CRTDefect/Directional Balance part of the SC2 package. Otherwise all $a_p \leq \eta$. Starting from the smallest remaining prime, add primes to the current layer until adding the next one would make the partial sum exceed 0.4. The completed layer has sum at most 0.4, and because the next summand is at most η, every non-final completed layer has sum greater than $0.4 - \eta$. This gives the claimed adaptive true-hit decomposition. □

Proposition 2.8 (Recorded conditional closure chain for the two-point secondary sieve)**.** *Using the three lemmas above, assume the following remaining input holds with the constants recorded in the secondary-sieve research note:*

1. *the I3-Core SC2/CRTDefect package controls the true-residual two-line near-crossing error, the odd-prime* 45° *main-synchronization peak, Single-Prime CRTDefect exits, the normalized* $q = 2$ *parity skeleton, the moment bounds* M_1, M_2, M_3*, and the endpoint Directional Balance used in the zero-mass and smoothing discharge lemma.*

Then the two-point secondary-sieve contradiction field has no remaining top-scale escape channel under the parameter choice

$$\alpha_0 = 0.75, \qquad c_0 = 1/2, \qquad \nu_j^{\mathrm{real}} \leq 0.4$$

for adaptive true-hit layers, subject to the same fixed-w convention used in the research manuscript.

Proof sketch and status. The argument is a conditional synthesis, not a final unconditional theorem. A top-scale covering excess is first decomposed into adaptive true-hit layers; any single-prime concentration is routed to the Single-Prime CRTDefect exit. The remaining excess is converted by a TCA linear bridge into a positive short-block load. The block load is forced into the SMC/SC2 near-crossing ledger. The 45° main synchronization is split into large-factor reuse, shared-complement factors, and good synchronized pairs; reuse and shared factors are excluded by short-window rigidity, while good pairs are sent to an auxiliary-prime Fourier CRT-defect. Directional balance excludes the Fourier defect unless the top-scale energy is already below the capacity threshold. Finally, the weighted Bonferroni zero-mass estimate supplies enough zero-hit block mass to make the capacity comparison incompatible.

The current audit is archived under `docs/archive/monograph-reviews/` and classifies this as a research proposition conditional on the single I3-Core package above. The fixed-order complexity input, zero-mass/smoothing discharge, and adaptive layering dichotomy have been inlined as the preceding lemmas. The fact that an unconditional proof would imply fixed-gap prime-pair consequences, including the twin-prime case when $w = 2$, is not a logical obstruction. It only marks the strength of the claim. The reason this chapter remains conditional is that I3-Core has not yet been inlined as an independently verified unconditional proof. □

Proposition 2.9 (Window-compressed conditional variant)**.** *Keep the same fixed even integer* $w > 0$ *and the same remaining I3-Core input as in the preceding proposition. Let* I *be any consecutive back-half window of* $H(P)$ *rows in the* $P \times P$ *matrix, so that* $|I| = H(P)P + O(P)$*. By the fixed-order complexity lemma, write*

$$Q_{\mathrm{eff}} \leq C_Q(\log P)^C.$$

For a prescribed middle-scale error budget $\varepsilon_{\mathrm{SC2}} > 0$*, the conditional row threshold supplied by the current ledger is*

$$H_{\min}^{\mathrm{cond}}(P; \varepsilon_{\mathrm{SC2}}) = \left\lceil \frac{C_Q}{\varepsilon_{\mathrm{SC2}}} P^{1/2} (\log P)^C \right\rceil.$$

Equivalently, one may use the asymptotic form

$$H(P) \geq H_{\min}^{\mathrm{asym}}(P) := \left\lceil P^{1/2}(\log P)^{C_*} \right\rceil, \qquad C_* > C.$$

Under either threshold, the same conditional contradiction-field chain gives an element

$$x \in I, \qquad p \nmid x(x-w) \quad (p \leq P).$$

When $x - w > 1$, both x and $x - w$ are primes.

Recorded scale check. The only scale-sensitive term in the recorded chain is the middle/high-scale $K = 2$ near-crossing error. With top layer $Y = P^{3/4}$ and $Q_{\mathrm{eff}} \leq (\log P)^C$, replacing the original back-half length $\asymp P^2$ by $|I| \asymp H(P)P$ changes the normalized error to

$$\frac{Q_{\mathrm{eff}} Y^2}{|I|} \ll \frac{P^{1/2}(\log P)^C}{H(P)}.$$

The displayed lower bound on $H(P)$ makes this at most $\varepsilon_{\mathrm{SC2}}$ in the explicit form, or $o(1)$ in the asymptotic form. Hence the constants in the TCA, SC2, zero-mass, finite-package, and smoothing ledgers are unchanged. The conclusion is therefore a conditional window-compressed version of the previous proposition. It becomes an unconditional prime-pair theorem only if I3-Core is independently proved and accepted. □

Remark 2.10 (Referee boundary)**.** The two-point secondary-sieve chapter is included to document the contradiction-field architecture and its parameter ledger. The window-compression audit is recorded in `docs/monograph/two-point-window-compression-and-unconditionality.md`. It must not be cited as an unconditional proof of prime pairs, the twin-prime conjecture, or Goldbach-type assertions unless I3-Core is independently proved and accepted.

Remark 2.11 (Current obstruction)**.** The current hard attack reduces I3-Core to a true-residual correlation problem. The local CRT and matrix rigidities do not by themselves exclude a model residual set concentrated on a new prime's bad residue class while still satisfying all previously imposed two-residue exclusions and short-window non-reuse constraints. Thus the remaining task is not another bookkeeping step but a genuine lower-bound/correlation theorem for the true residual set in new prime moduli.

Remark 2.12 (Latest TRC-1G reduction)**.** The single-prime part of the remaining TRC input has been sharpened to a heavy-atom prohibition. If more than 40% of the true residual mass lies in the two bad classes $0, w \pmod p$ for some new prime $p > Y$, then Parseval gives a non-zero Fourier coefficient modulo p of size $> 1/5$ after normalization. Equivalently, the three long-direction difference families $0, \pm w \pmod p$ carry a fixed positive proportion of the true-residual pair energy. The existing matrix rigidities remove the short-direction, 45°-locked, reuse/shared-complement, and endpoint pieces. The remaining unproved input is therefore a new-modulus long-direction balance theorem for the true residual set, not a finite-template or local CRT bookkeeping issue.

Remark 2.13 (Total large-incidence reduction)**.** The pointwise new-modulus balance can be weakened further. It is enough to prove the total large-incidence inequality

$$\sum_{P^\alpha < p \leq P} \#\{x \in U_{P^\alpha}(I) : p \mid x(x-w)\} < |U_{P^\alpha}(I)|.$$

Indeed, a point not counted on the left is unsieved by every prime up to P, and in the P^2 matrix range this forces both x and $x - w$ to be prime. The CRT model predicts the normalized left side to be $2\log(1/\alpha) + o(1)$; for $\alpha = 3/4$ this is $2\log(4/3) = 0.57536\ldots$, leaving margin $0.42463\ldots$. The unproved part is the true-residual total-incidence estimate with an error below that margin.

Remark 2.14 (RB–TLI audit boundary)**.** The direct attempt to prove the total large-incidence estimate separates cleanly into a numerator and a denominator. The numerator is an upper-sieve problem: after writing $x = pm$ or $x = w + pm$, the variable m again avoids two residue classes modulo each old prime, giving an upper bound of shape

$$A_p \le 2C_+|I|V_2(P^\alpha)/p + R_p.$$

The denominator requires a model-level lower bound $|U_{P^\alpha}(I)| \ge c_-|I|V_2(P^\alpha)$. For $\alpha = 3/4$, this route would need $C_+/c_- < 1/(2\log(4/3)) = 1.738\ldots$. This relative Buchstab total-incidence input is the remaining unproved prime-pair-strength statement.

Remark 2.15 (Local collapse when $q \mid w$)**.** If an old odd prime q divides the fixed even difference w, then the two forbidden classes $0, w \pmod q$ coalesce into a single forbidden class. The local density is therefore $1-1/q$ instead of $1-2/q$, giving the singular local gain $(q-1)/(q-2)$. This increases the absolute number of admissible residual points and weakens the small-prime synchronization peaks. In the relative RB–TLI ratio, however, the same local factor multiplies both $|U_Y|$ and the model numerator $A_p \sim 2|I|V_w(Y)/p$ for all new primes $p > Y > |w|$. Thus the main relative constant remains $2\log(1/\alpha)$; the collapse improves finite constants but does not by itself close the hardest case $w = 2$.

Remark 2.16 (Buchstab semiprime transition for $w = 2$)**.** The latest numerical audit of the hardest case $w = 2$ shows that the naive constant $2\log(1/\alpha)$ is not the correct conditional main term after the old primes have been sieved. For $\alpha > 2/3$, any new-prime hit $p \mid x$ or $p \mid x-2$ with $Y = P^\alpha < p \le P$ forces the complementary quotient to be prime, because it is Y-rough and smaller than Y^2. Thus the large-incidence layer is a Buchstab semiprime transition. The corresponding two-sided main constant is

$$K(\alpha) = \frac{2\log((2-\alpha)/\alpha)}{1+\log((2-\alpha)/\alpha)}.$$

Exact scans recorded in `docs/rb-tli-w2-scan-large.md` give, for example, $E_UD = 0.691696$ at $P = 10007, \alpha = 3/4$, compared with $K(3/4) = 0.676220$. The remaining target is therefore a two-point stability theorem for this Buchstab semiprime transition, not the earlier naive incidence model.

Remark 2.17 (BST error decomposition)**.** The refined scan decomposes the remaining error. At $P = 10007, \alpha = 3/4$, the shell contributions in the five exponent intervals between 0.75 and 1 track the Buchstab shell constants, and the covariance between the x-side and $(x-2)$-side large-incidence counts is approximately $-9.3\cdot 10^{-5}$. The zero-incidence set is independent of the intermediate cutoff α, as it is exactly the set unsieved by all primes up to P. Thus the remaining hard input has been reduced to a shifted biprime roughness estimate of the form

$$\#\{p, m : pm \in I,\ pm-2 \in U_Y\} \le (B(\alpha) + o(1))|U_Y|.$$

This is the current BST–2 obstruction.

Remark 2.18 (BMD reduction)**.** The BST–2 obstruction can be written as a one-dimensional multiplicative sieve on biprime variables. After setting $x = pm$, the condition $pm-2$ is unsieved by old primes is, for each $q \le Y$,

$$pm \not\equiv 2 \pmod q.$$

The prime $q = 2$ is peeled off deterministically: for odd primes p, m, the integer $pm-2$ is odd. Thus the multiplicative sieve is taken over odd q. For each odd $q \le Y$, since $p, m > Y \ge q$, the forbidden curve has $(q-1)$ points inside $(\mathbb{F}_q^\times)^2$, out of $(q-1)^2$ possible non-zero pairs. The local forbidden density is therefore $1/(q-1)$. The remaining unproved estimate is a biprime multiplicative dispersion bound: weighted sums of the errors in the congruences $pm \equiv 2 \pmod d$, with odd squarefree d, must be $o(|U_Y|)$ for the Buchstab/Rosser weights used in the transition. This is the current BMD input.

Remark 2.19 (Row-column input for BMD)**.** If the prime-matrix row/column theorem is accepted as a conditional input, it can be used only to remove zero-section degeneracies in complete CRT blocks: entire empty rows, columns, or unit residue cells are then not allowed to masquerade as the main BMD error. It does not by itself prove BMD, because BMD is a signed distribution statement. For odd d, expanding

$$1_{pm\equiv 2 \pmod d} = \frac{1}{\varphi(d)} + \frac{1}{\varphi(d)} \sum_{\chi\neq\chi_0} \overline{\chi}(2)\chi(p)\chi(m)$$

shows that the remaining obstruction is a bilinear character-dispersion estimate

$$\sum_{d\le D} \frac{|\lambda_d|}{\varphi(d)} \sum_{\chi\neq\chi_0} \left| \sum_{Y<p\le P} \chi(p) \sum_{\substack{m\in J_p \\ m \text{ prime}}} \chi(m) \right| = o(|U_Y|).$$

Thus the row/column input gives a useful BMD-Zero exclusion, but the genuine remaining hard point is BMD-Char.

Remark 2.20 (Direct BMD attack via an E_2 Bombieri–Vinogradov input)**.** Set $N \asymp P^2$ and

$$a_n = \#\{(p,m) : Y < p \le P,\ m \text{ prime},\ n = pm,\ n \in I\}.$$

For $\alpha > 2/3$ the multiplicity of a_n is bounded. After the deterministic modulus 2 has been peeled off, the BMD remainder for odd squarefree d is exactly the arithmetic-progression discrepancy of this restricted two-prime convolution:

$$R_d = \sum_{\substack{n\in I \\ n\equiv 2 \pmod d}} a_n - \frac{1}{\varphi(d)} \sum_{\substack{n\in I \\ (n,d)=1}} a_n.$$

Thus BMD follows from the standard Bombieri–Vinogradov theorem for restricted E_2 sequences at level

$$d \le P(\log P)^{-B} = N^{1/2}(\log N)^{-B+O(1)}.$$

Indeed, if

$$\sum_{d\le P/\log^B P} \max_{(a,d)=1} |R_d(a)| \ll_A N(\log N)^{-A},$$

then every Rosser/Buchstab weighted remainder is $o(|U_Y|)$ since $|U_Y| \asymp N/(\log P)^2$. The condition $\alpha < 1$ ensures $Y < P/\log^B P$ for large P, so all one-prime forbidden curves $q \le Y$ are included in the available distribution range. Therefore BMD is closed if this standard E_2 BV input is cited; a fully self-contained manuscript must instead prove that input in an appendix.

Remark 2.21 (Self-contained BV–E_2 obligation)**.** The local BV–E_2 appendix records the exact self-contained obligation. A direct multiplicative large-sieve estimate is insufficient: in the balanced dyadic block $p \sim P_1$, $m \sim M_1$, with $P_1 \asymp M_1 \asymp P$ and $Q \asymp P/\log^B P$, Cauchy plus the large sieve gives size roughly $P^3/\log^{2B} P$, while the required Bombieri–Vinogradov error is $P^2/\log^A P$. Thus the only genuinely deep remaining self-contained input is a Type-II dispersion/Kloosterman-cancellation estimate for the balanced convolution blocks. If that standard dispersion theorem is cited, the BMD step is closed as an external-input theorem; if not, this is the next appendix that must be proved line by line.

Remark 2.22 (Weighted BE2–3K core)**.** BMD does not require the full $\max_a$ Bombieri–Vinogradov statement. It only needs the fixed class $2 \bmod d$ against the well-factorable Rosser/Buchstab weights. After Cauchy, the self-contained problem is a variance bound for

$$T_r = \sum_{d \le Q} \lambda_d \left(\sum_{s \equiv 2r^{-1} \bmod d} \beta_s - \frac{1}{\varphi(d)} \sum_{(s,d)=1} \beta_s \right).$$

Opening $\sum_r |T_r|^2$, the off-diagonal conditions

$$rs_1 \equiv 2 \pmod{d_1}, \qquad rs_2 \equiv 2 \pmod{d_2}$$

give, by CRT and Fourier expansion of the interval condition, the Kloosterman phase

$$e\left(-\frac{2h\overline{s_1}\,\overline{d_2}}{d_1} - \frac{2h\overline{s_2}\,\overline{d_1}}{d_2} \right).$$

Thus the true no-black-box core is a weighted bilinear Kloosterman dispersion estimate, denoted BE2–3K in the notes. Proving BE2–3K gives BE2–3, then weighted BV–E_2, then BMD.

Remark 2.23 (Referee status of the no-black-box claim)**.** At the current stage the manuscript is not yet fully no-black-box. The BMD obstruction has been reduced to the single deep BE2–3K Kloosterman-dispersion estimate. If a BFI/Deshouillers–Iwaniec/Kuznetsov type mean-value theorem is cited, BMD is closed as an external-input theorem. If no external input is allowed, BE2–3K remains the unique missing line-by-line proof obligation.

Remark 2.24 (Further reduction of BE2–3K)**.** The BE2–3K core can be narrowed further. Pointwise Weil bounds for individual incomplete reciprocal sums give only single-modulus square-root cancellation and do not provide the arbitrary logarithmic saving required after summing over the d_1, d_2, h families. The common-divisor strata $(d_1, d_2) > 1$ are not the main obstruction: compatibility forces $s_1 \equiv s_2$ modulo the common divisor and yields only polylogarithmic loss. The remaining genuine input is a windowed Kloosterman spectral large-sieve estimate, denoted KLS-window in the notes. Its role is precisely

$$\text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{weighted BV–}E_2 \Rightarrow \text{BMD}.$$

Remark 2.25 (External-theorem closure of KLS-window)**.** The KLS-window input is tied to two classical external sources: the Deshouillers–Iwaniec spectral Kloosterman large-sieve technology, as developed in *Kloosterman sums and Fourier coefficients of cusp forms*, Invent. Math. 70(2), 219–288 (1982), and the Bombieri–Friedlander–Iwaniec dispersion framework with well-factorable weights, as in *Primes in Arithmetic Progressions to Large Moduli. II*, Math. Ann. 277, 361–394 (1987). With these external theorems allowed, the analytic chain closes as

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{weighted BV–}E_2 \Rightarrow \text{BMD}.$$

This is an external-deep-theorem closure, not a fully self-contained reproof of Kuznetsov/dispersion theory.

Chapter 3

Zeta-Zero Contradiction Fields

Assume an off-critical zero $\rho = \beta + i\gamma$ with $\beta > 1/2$. The RH contradiction-field manuscript transfers the resulting smooth prime anomaly into a CRT rough-composite ledger, decomposes it into sparse/dense/tail/internal/global exits, and proves that no named escape channel remains under the recorded local theorems and restricted external inputs.

Theorem 3.1 (RH contradiction-field synthesis, recorded form)**.** *Under the numbered local theorems, controlled-exit lemmas, threshold hierarchy, and restricted external inputs recorded in `paper/rh-proof/rh-contradiction-field.tex`, an off-critical zero has no free escape channel in the contradiction-field event graph.*

Reference. See the RH manuscript in `paper/rh-proof/` and the final audit in `docs/`. The submission warning in the RH manuscript remains in force until independent line-by-line referee verification and final monograph page-number checks are completed. □

Chapter 4

Unified Dependency Map

The two programs share the same logic:

$$\text{counterexample} \Rightarrow \text{structured covering field} \Rightarrow \text{named exits} \Rightarrow \text{capacity/no-cycle contradiction.}$$

The prime-matrix program is more geometric and finite-CRT in nature; the zeta-zero program is analytic and scale-asymptotic. The monograph treats both as projections of the same contradiction-field method.

Layer	Prime-matrix projection	RH projection
Baseline	CRT nonzero class balance	smoothed CRT candidate baseline
Local rigidity	large-factor non-reuse, diagonal locks	sparse/dense/tail routing
Capacity	anchor coverage capacity	projection and square-function capacity
No-cycle	recursive peeling and shell migration	GEE transfer/no-cycle ledger
External inputs	finite verification and Structured-EHPD	explicit formula, KL, Vaaler, BG/Baker, sieve, Vaughan
Final status	reduction package, not final unconditional theorem	verification package with submission warning

Chapter 5

Directory and Theory System

The detailed working directory and theory-system map is maintained in

`docs/monograph/combined-monograph-directory-and-theory-system.md`.

This chapter records the concise version used by the merged monograph.

5.1 Proposed Book Structure

Part	Contents	Status
I	Unified contradiction-field method: entrance, local rigidity, exits, capacity, no-cycle potential	Methodological framework
II	Prime-matrix row/column program: CRT skeleton, A/B reduction, Structured-EHPD interface	Reduction package; not final theorem
III	Two-point secondary sieve: TLI, BST, BMD, WBE2, BE2–3K, KLS-window	External-deep-theorem closure for BMD
IV	Zeta-zero contradiction field: explicit-formula entrance and controlled exits	Verification package; not final RH theorem
V	Dependency map, referee checklist, optimization routes	Audit and future work

5.2 Latest Two-Point Chain

For the $w = 2$ hardest two-point branch, the current analytic chain is

$$\mathrm{DI{+}BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{WBE2} \Rightarrow \text{BMD} \Rightarrow \text{BST–2} \Rightarrow \text{BST} \Rightarrow \text{TLI}.$$

The chain is closed only in the external-deep-theorem sense: DI and BFI are cited as classical analytic inputs. A fully self-contained version would require reproving the spectral Kloosterman large-sieve and dispersion machinery.

5.3 Prime-Matrix Low-Hole Closure Package

The latest prime-matrix update closes the low-hole bucket submodule as a finite-certificate plus explicit-external-theorem package. The verification is split into five non-overlapping ranges:

$$
\begin{aligned}
13 \le P < 61 &: \text{ low-range final certificate,}\\
61 \le P \le 103 &: \text{ narrow-band collision-energy certificate,}\\
107 \le P \le 229 &: \text{ exact } P/5 \text{ splitting recursion,}\\
233 \le P \le 13207 &: \text{ integer cross-multiplied product certificate,}\\
P \ge 13208 &: \text{ Rosser–Schoenfeld explicit constants.}
\end{aligned}
$$

The low-range certificate checks 15414 zero buckets: 15282 close by whole-set Hall deficit, 108 close by bridge-critical deletion, and the remaining 24 exceptional phases at $P = 41$ close by exact finite set-cover dynamic programming. The narrow band $61 \le P \le 103$ covers 23100 low phases by fixed ascending high-prime residue blocks. The two finite tail certificates cover 23 and 1520 prime rows respectively, the latter by exact integer cross multiplication rather than floating-point comparison.

For the infinite tail, Rosser–Schoenfeld Corollary 1, formulas (3.5)–(3.6), supplies the prime-counting bounds, while Theorem 7, formula (3.26), supplies the Mertens product bound. Since $P \ge 13208$ implies $\lfloor P/5 \rfloor \ge 2641$, the factor $1 + 1/(2\log^2\lfloor P/5\rfloor)$ is below 1.009, hence the manuscript's conservative 1.03 Mertens constant is admissible.

This closes the BPN low-hole bucket interface in the stated finite-certificate/external-theorem sense. It does not by itself promote the whole prime-matrix row/column program to a final theorem, because the remaining global exits still require the separately named PDEC/SAE and Rankin-ledger acceptance obligations.

5.4 Prime-Matrix WSH-Hall Phase Certificate

The latest row-program compression keeps the endpoint status honest. The terminal RCI branch has been reduced to the comparison between no-tail primes and balanced two-tail semiprimes. The next local interface is

$$\text{RCI/PDEC} \Rightarrow \text{Distributed-RCI} \Rightarrow \text{WSH-Hall/PDEC}.$$

Here WSH-Hall means that every balanced two-tail semiprime in a row can be injectively matched to a same-row prime through an offset allowed by the small-prime wheel. If such a Hall matching fails, the failing interval must be routed to a named exit: persistent endpoint CRT defect, tail-anchor concentration, or endpoint prime deficit.

A finite phase certificate is archived in

`docs/monograph/prime-matrix-wsh-hall-phase-certificate.md`.

For $17 \le p \le 2000$, $y = \lfloor p/e \rfloor$, and wheel primes $2, 3, 5, 7, 11, 13$, it scans 215074 rows containing balanced two-tail semiprimes. It finds no Hall matching failure, no failure under the candidate radius $3\log^2 q$, maximum minimal matching radius 132, maximum ratio $R/\log^2 q = 2.3771082468335174$, and minimum margin $\#\text{primes} - \#\text{semiprimes} = 1$. The certificate rows list semiprime labels, matched primes, offsets, fixed-offset loads, and wheel-phase loads; each matched offset is checked against the congruence condition $d \not\equiv -b \pmod r$.

This is a computational-certificate update, not a global theorem. The remaining proof obligation is either a uniform WSH-Hall theorem or a proof that every Hall defect necessarily triggers PDEC, Tail-anchor, or Endpoint deficit.

5.5 WSH-Hall Defect Trichotomy

The corresponding defect audit is archived in

`docs/monograph/prime-matrix-wsh-hall-defect-trichotomy.md`.

It converts a possible Hall failure into a consecutive block B_0 of balanced two-tail semiprimes with

$$|N_R(B_0)| < |B_0|.$$

The same block is then examined through five projections: endpoint prime deficit, tail-factor load, small-wheel residue load, fixed-offset capacity, and the terminal mirror $n \mapsto q^2 - n$. The resulting target is the named interface

$$\text{WSH-Expansion-or-Defect}.$$

If Tail-anchor and fixed-offset/PDEC thresholds do not fire, one must prove the expansion inequality

$$|N_R(B_0)| \geq |B_0|$$

for $R = C \log^2 q$; otherwise the failing block must be absorbed by Endpoint/PDEC. A forced-radius audit on the 44 tight certificate rows produces 124 explicit Hall defects and records their tail, phase, offset, and mirror data. This audit fixes the certificate fields and exit geometry; it is not a substitute for the uniform expansion theorem.

5.6 WSH Expansion Margin Ledger

The expansion term itself is audited in

`docs/monograph/prime-matrix-wsh-expansion-margin-ledger.md`.

For $17 \leq p \leq 2000$, $R = \lceil 3 \log^2 q \rceil$, and the same $y = \lfloor p/e \rfloor$, the audit scans every consecutive balanced-semiprime block in every relevant row. It checks 8440419 blocks across 215074 rows and finds

$$\min_B \bigl(|N_R(B)| - |B|\bigr) = 1,$$

with no zero-surplus block. Only seven blocks have surplus at most 1.

This strengthens the finite certificate from “a matching exists” to “every interval-Hall block has positive margin” in the audited range. It still does not prove the uniform theorem. The remaining target is:

$$\text{WSH Positive Expansion or Named Defect}.$$

After Tail-anchor, fixed-offset/PDEC, and endpoint-mirror deficit exits are removed, one must prove the positive expansion inequality for all formal blocks.

5.7 Short Critical Block Reduction

The short-critical-block audit is archived in

`docs/monograph/prime-matrix-wsh-short-critical-block-reduction.md`.

It reruns the expansion-margin audit with tight surplus threshold 2. Among the same 8440419 consecutive semiprime blocks, only 45 have surplus at most 2, and every such block has at most three semiprimes. The distribution is:

surplus	$\lvert B\rvert$	count
1	1	4
1	2	3
2	1	24
2	2	10
2	3	4

This suggests the next reduction:

$$\text{SCB–1: } |B| \geq 4 \Rightarrow \text{positive expansion or named defect,}$$

$$\text{SCB–2: } |B| \leq 3 \Rightarrow \text{local exclusion or named defect.}$$

This is still finite evidence and a proof target, not a theorem. It is useful because SCB–2 is a small local problem involving only one, two, or three two-tail semiprimes, where tail labels, fixed offsets, wheel residues, and endpoint mirrors can be written explicitly.

5.8 SCB–2 Local Routing Certificate

The SCB–2 local certificate is archived in

`docs/monograph/prime-matrix-wsh-scb2-local-exclusion-template.md`.

It enumerates all consecutive blocks with $|B| \leq 3$ for $17 \leq p \leq 2000$. The block counts are

$\lvert B\rvert$	count	$\min(\lvert N_R(B)\rvert - \lvert B\rvert)$
1	1496400	1
2	1281326	1
3	1088870	2

There are no negative-surplus or zero-surplus short blocks. The local routing statement is:

$$|B| \leq 3, \quad |N_R(B)| < |B| \Rightarrow \text{Endpoint/PDEC deficit,}$$

with additional Tail-repeat and fixed-offset-full-load checks on tight two- and three-point blocks. Thus SCB–2 is closed as a routing lemma modulo the still-open Endpoint/PDEC exclusion.

5.9 SCB–1 Long-Block Certificate

The complementary long-block certificate is archived in

`docs/monograph/prime-matrix-wsh-scb1-long-block-expansion-template.md`.

It enumerates all consecutive balanced two-tail semiprime blocks with $|B| \geq 4$ for $17 \leq p \leq 2000$ and the same radius $R = \lceil 3 \log^2 q \rceil$. The audit checks 4573823 long blocks and finds

$$\min_{|B| \geq 4} \big(|N_R(B)| - |B|\big) = 3,$$

with no negative-surplus or zero-surplus long block. The four tight long blocks have sizes 4 or 5, and every tight block carries both Endpoint-margin and fixed-offset-full-load labels.

This is a finite certificate, not the uniform SCB–1 theorem. Its proof-theoretic role is to isolate the next target:

$$\text{long-block positive expansion} \quad \text{or} \quad \text{fixed-offset/PDEC absorption.}$$

Together with SCB–2, the present WSH chain is therefore:

$$\text{SCB–2 routing modulo Endpoint/PDEC}, \qquad \text{SCB–1 finite long-block margin},$$

The remaining global obstructions are fixed-offset/PDEC absorption and Endpoint/PDEC exclusion.

5.10 Fixed-Offset/PDEC Absorption

The fixed-offset absorption interface is archived in

`docs/monograph/prime-matrix-wsh-fixed-offset-pdec-absorption.md`.

It proves the no-third-escape lemma behind the fixed-offset-full-load label. If $p < q$ are consecutive primes and $1 < n < q^2$ is composite, then n has a prime factor at most p. Hence, if a fixed offset d is allowed by the wheel $2, 3, 5, 7, 11, 13$ for every $b \in B$, every missing candidate $b + d$ is explained by some factor in $(13, p]$.

The finite ledger

`docs/monograph/prime-matrix-wsh-fixed-offset-pdec-ledger.md`

checks the tight SCB–1 long blocks. It finds 11 full-load offset rows and 47 candidates, among which 15 are prime and 32 are missing. Every missing candidate has an explaining factor at most p; there is no unexplained candidate.

Thus fixed-offset-full-load is no longer an unnamed escape. It routes to expansion, Tail-anchor, PDEC/low-mod CRTDefect, or SAE/Endpoint. The still-open global target is

$$\text{FO-PDEC: distributed explaining factors} \Rightarrow \text{persistent PDEC or sparse SAE/Endpoint.}$$

5.11 FO-PDEC Low-Mod Equations

The low-mod equation audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-hard-attack.md`.

For every missing candidate $n = b + d = (r - 1)q + c$ and every explaining factor ℓ, the exact CRT equation is

$$r \equiv 1 - cq^{-1} \pmod{\ell}.$$

If $b = uv$ is the corresponding two-tail semiprime, the same missing candidate also satisfies

$$uv + d \equiv 0 \pmod{\ell}.$$

The finite audit checks 43 such low-mod equations in the tight fixed-offset ledger. There are no CRT-equation failures and no bilinear-equation failures.

This closes the equation layer of FO-PDEC. It does not close the global theorem. The remaining target is the quantitative inequality

$$\text{low-mod defect energy} \geq \text{PDEC threshold},$$

unless the non-persistent part is absorbed by SAE/Endpoint.

5.12 FO-PDEC Energy Lower Bound

The energy ledger is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-energy-lemma.md`.

For a multiset E of explaining equations, let t_ℓ be the number of equations using ℓ and let s_ℓ be the number of occupied row residues modulo ℓ. Define

$$\mathcal{E}_\ell(E) = t_\ell - \frac{s_\ell t_\ell}{\ell}, \qquad \mathcal{E}(E) = \sum_\ell \mathcal{E}_\ell(E).$$

For every $0 < \theta < 1$, either some explaining factor has high load $t_\ell > \theta\ell$, which is a Tail/PDEC exit, or

$$\mathcal{E}(E) \geq (1-\theta)|E|.$$

Thus FO-PDEC now produces low-mod energy unconditionally, modulo the named high-load exit.

The finite ledger with $\theta = 0.25$ gives global energy per equation

$$0.9547817296210457$$

and no high-load factor. The remaining obstruction is not energy production; it is the final threshold comparison on the same bad-window set:

$$U_{\mathrm{CRT}}(E) < \mathcal{E}(E) \quad \text{or} \quad L_{\mathrm{PDEC}}(E) \leq (1-\theta)|E|.$$

5.13 FO-PDEC Explicit Threshold

The explicit threshold ledger is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-threshold-comparison.md`.

For each explaining factor ℓ, it computes the nonzero Fourier threshold

$$M_\ell(E) = \max_{1 \leq h < \ell} \left| \sum_{\rho \bmod \ell} g_\ell(\rho) e^{2\pi i h\rho/\ell} \right|.$$

The current best projection is

$$\ell = 199, \qquad h = 95, \qquad M_\ell(E) = 3.959247567099438.$$

Thus the remaining task is now concrete:

$$U_{\mathrm{CRT},199} < 3.959247567099438$$

for the same vector g_{199}. Otherwise this projection must be absorbed by SAE/Endpoint.

5.14 FO-PDEC Dual-Cluster Obstruction

The dual-cluster audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-dual-cluster-route.md`.

For the best projection, the mass is 4 and the Fourier threshold is 3.959247567099438, so the trivial mass bound misses closure by only

$$0.04075243290056196.$$

The best frequency $h = 95$ sends the support to a short dual arc of length 11 modulo 199:

$$40, 126, 61 \mapsto 19, 30, 24.$$

Thus the last obstruction is a concrete dual-cluster exclusion problem: this short dual arc must be ruled out by a PDEC dual certificate, or else absorbed as a non-persistent SAE/Endpoint event.

5.15 FO-PDEC Multiplicity Legitimacy

The primitive-cluster audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-multiplicity-legitimacy.md`.

It checks whether the best projection counts independent bad-window equations or repeated physical candidates. Under the equation or block-local convention, the projection has mass 4 and Fourier value

$$3.959247567099438.$$

Under physical deduplication, it has mass 2 and Fourier value

$$1.9699193446802263.$$

Thus the final PDEC comparison must first fix the multiplicity convention. The lower bound and the upper bound must be computed on the same multiset. Otherwise the argument must switch to the primitive threshold or absorb the repeated events by SAE/Endpoint.

5.16 FO-PDEC Formal Unit Consistency

The formal-unit audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-formal-unit-route.md`.

It checks a stricter requirement: the PDEC vector $g(t)$ must come from one formal bad-window set or multiset with one phase map. The current strong value

$$3.959247567099438$$

is obtained by pooling the finite certificate library. After splitting by formal unit, the audit gives

convention	best Fourier value
global library	3.959247567099438
layer deduplication	2.9698366905785227
physical deduplication	1.9997507790353146
single coordinate-deduplicated formal unit	1.0.

Thus the strong threshold cannot yet be used as a single-branch PDEC lower bound. One must prove FormalUnit-Stitching for the cross-q events and NestedBlock-Independence for exact nested-coordinate duplicates, or route the repeated events to SAE/Endpoint. This is now the narrowest FO-PDEC obstruction before any global unconditional conclusion can be claimed.

5.17 FO-PDEC Branch Separation

The branch-separation theorem and stitching audit are archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-branch-separation-theorem.md`.

They separate two issues. First, events from different matrix widths q cannot be pooled into one PDEC vector without a cross-q persistence theorem. Second, two occurrences with the same

$$(q, row, candidate\_row, column, offset, n, \ell, \rho)$$

give the same CRT equation and must be quotiented unless a weighted Hall-dual independence row is supplied. In the present finite ledger there are 7 exact nested-coordinate duplicates and 9 cross-level physical reuses. After coordinate deduplication inside a single formal unit, the best Fourier value recorded by the audit is only 1.0. Therefore the next narrow obstruction is Weighted Hall Dual Independence, or else SAE/Endpoint absorption of the exact duplicates.

5.18 Claim Status Legend

The monograph uses the following status labels.

Label	Meaning
Proved-in-text	The statement is proved inside the manuscript under the stated hypotheses.
Reduction-closed	The statement has been reduced to a smaller named interface.
External-theorem closed	The statement is closed after citing explicitly named external theorems.
Computational-certificate	The statement is closed over a finite range or certified submodule by reproducible scripts and archived outputs.
Referee-block	A proof package exists, but it still requires line-by-line referee verification.
Not claimed	The statement is not asserted as a theorem of the manuscript.

The status discipline and promotion rules are maintained in

`docs/monograph/claim-status-discipline.md`.

The manuscript provides theorem-like environments with matching names for future editing. A statement may be promoted only when its required proof, reduction endpoint, external theorem template, or certificate package is present.

5.19 External Theorem Index

The external-input index is maintained in

`docs/monograph/external-theorem-index.md`.

For the prime-matrix low-hole tail, the current explicit package is Rosser–Schoenfeld 1962: Corollary 1, formulas (3.5)–(3.6), supplies the required $\pi(x)$ bounds, and Theorem 7, formula (3.26), supplies the required Mertens product bound. These constants close the $P \geq 13208$ tail in the external-theorem sense once the finite certificates below that threshold are accepted.

For the two-point sieve, the critical external package is DI plus BFI. The index records the required checks for the KLS-window input: Kloosterman phase normalization, modulus range, frequency range, inverse-variable support, well-factorable weights, gcd strata, smoothing and sawtooth losses, and the final choice of the logarithmic-loss parameter $B(A)$. The standalone adaptation template is maintained in

`docs/monograph/kls-window-di-bfi-adaptation-template.md`.

It closes the BMD distribution input only in the external-deep-theorem sense; the separate BMD-to-TLI transfer remains a hard obligation.

5.20 Critical Chain Audit

The detailed chain audit is maintained in

`docs/monograph/key-proof-chain-optimization-audit.md`.

Its main purpose is to prevent status mixing. Prime-matrix existence rigidity may support zero-section exclusion, but it does not replace signed spectral distribution estimates such as BMD-Char. Numerical experiments may guide constants and structures, but they are not proof inputs. The RH branch remains a verification package and is not claimed as an unconditional proof.

5.21 Optimization Priorities

1. Separate theorem environments for proved-in-text, reduction-closed, external-theorem closed, and referee-block statements.

2. Replace the overly strong $\max_a$ BV–E_2 formulation by the precise WBE2 well-factorable weighted statement in the main line.

3. Rewrite KLS-window directly in the notation of the cited DI/BFI theorems to reduce translation risk.

4. Compress the prime-matrix program around the single Structured-EHPD exclusion interface.

5. Preserve the RH warning language until every controlled exit has independent referee verification.

5.22 Pre-External-Referee Hard Obligations

The current hard-obligation ledger is maintained in

`docs/monograph/pre-external-referee-remaining-obligations.md`.

It records the author-side status of every terminal interface that cannot honestly be promoted to a final theorem without further proof or independent verification. The decisive interfaces are:

Program	Hard interface	Required closure before theorem promotion
Prime matrix	Structured-EHPD / D-structure exclusion	Prove the exclusion theorem directly, or replace it by a fully proved RHI-type large-prime incidence inequality.
Prime matrix	PDEC / SAE / Rankin exits	Prove persistent endpoint CRT defect exclusion, local isolated-escape exclusion, and complete the formal Rankin certificate ledger.
Prime matrix	RRD / OSPC / Selberg-uniform constants	Put all constants in one convention and prove each budget item over the full asymptotic range, not only in sampled windows.
Two-point sieve	I3-Core true-residual package	Prove true-residual new-modulus balance, TCA, SC2, 45° synchronization peak control, Directional Balance, Zero Mass, and endpoint smoothing.
Two-point sieve	DI/BFI to KLS-window adaptation	State the external theorems in their standard variables and verify phase, modulus, frequency, weights, gcd strata, and logarithmic-loss absorption.
Two-point sieve	BMD to TLI transfer	Prove the Buchstab transfer and total-large-incidence comparison without using a hidden prime-pair lower bound.
RH branch	controlled exits	Prove every sparse, dense, tail, internal, analytic, and global exit under a single load convention.

For the RRD/OSPC constants, the current acceptance matrix is maintained in

`docs/monograph/h5-rrd-ospc-proof-obligation-matrix.md`.

It splits the remaining constant obligation into six independently checkable inequalities and does not claim that those inequalities have already been proved. The first item, RRD-low routing, is recorded in

`docs/monograph/h5-1-rrd-low-exit-theorem.md`.

It proves that an RRD-low excess enters either OSPC* or a weighted CRT-defect exit; excluding those exits remains part of the PDEC/SAE and tail-anchor obligations. The absorption of these two exits into the unified PDEC-or-SAE interface is recorded in

`docs/monograph/h5-4-ospc-weighted-crtdefect-absorption.md`.

This absorption is not an exclusion theorem; the required PDEC and SAE certificates remain open hard obligations. The PDEC certificate format itself is now recorded in

`docs/monograph/h4-pdec-certificate-template.md`.

It fixes the data, Fourier lower bound, admissible CRT upper-bound certificates, and failure-routing rules for persistent defects. This is a certificate-format closure only; the actual PDEC certificates for the formal bad-window families remain to be supplied. The first source lemmas for admissible PDEC constraint rows are recorded in

`docs/monograph/h4-pdec-constraint-source-lemmas.md`.

They justify mass, phase-capacity, mirror-pair, low-hole-bucket, and conditional-routing rows when their stated hypotheses hold. They do not by themselves provide the full constraint table or the final dual upper bound. The first admissibility table for those rows is recorded in

`docs/monograph/h4-pdec-admissible-constraint-table.md`.

It separates tautological, finite-certificate, symbolic, conditional-routing, still-unproved, and rejected rows. In particular, full-cycle CRT balance is not allowed to control an arbitrary bad-window subset. The column-cap source rule is recorded in

`docs/monograph/h4-pdec-column-cap-source-lemma.md`.

It allows column information to enter only through finite column-projection capacity, symbolic column-capacity theorems, or conditional routing to ColumnRadius/ColumnCRT/Tail-anchor exits. The coefficient ledger for those rows remains open. The first coefficient ledger is recorded in

`docs/monograph/h4-pdec-column-cap-coefficient-ledger.md`.

It registers finite column-witness, RCI/CDB, LHB column-residue, and conditional ColumnDefect rows. Most rows still require machine-readable phase blocks before they can become actual rows of the dual certificate matrix. The first materialized phase-block file is

`docs/monograph/h4-pdec-lhb-column-phase-blocks.json`.

It gives finite $Q = 2310$ LHB column rows for $13 \le p \le 47$. Empty anomaly blocks are machine-readable finite rows; diagnostic support blocks still require a projection-capacity proof before entering the final dual system. The support-to-capacity transfer conditions are recorded in

`docs/monograph/h4-pdec-lhb-support-to-capacity-transfer.md`.

They make explicit that support size is not a multiplicity bound for the persistent count vector $g(t)$; a phase-indicator reduction, a multiplicity cap, or an allowed-set projection bound is still required. The multiplicity route is made explicit in

`docs/monograph/h4-pdec-lhb-multiplicity-cap-route.md`.

It states the precise acceptance contract: for the same (p, Q, S, τ) one must prove $g(t) \leq M(t)$, and only then may a diagnostic block C be promoted to the capacity row $\sum_{t \in C} g(t) \leq \sum_{t \in C} M(t)$. The first finite multiplicity-cap certificate is

`docs/monograph/h4-pdec-lhb-multiplicity-cap-certificate.json`.

It takes $M(t)$ to be the high-layer CRT completion count $C_P(t; Q)$ and verifies, for $Q = 2310$ and $p = 13, 17, 19, 23, 29, 31, 37, 43, 47$, that the WHOLEDEF and BRIDGED support blocks have $\sum_{t \in C} M(t) = 0$. This is an allowed-set projection certificate: it can be used in the LHB branch after proving $S \subset Z_{\mathrm{LHB}}(p, Q)$, but it is not by itself a global PDEC exclusion theorem. The attachment lemma for that branch is

`docs/monograph/h4-pdec-lhb-attachment-lemma.md`.

It proves that an LHB-typed full-row bad window is indeed an element of $Z_{\mathrm{LHB}}(p, Q)$. Thus the remaining global task is no longer this attachment step, but the classification step: every formal PDEC bad window must either be LHB-typed or be routed to a named SAE, ColumnCRT/ColumnRadius, TailAnchor, or Rankin exit. The classification step is recorded in

`docs/monograph/h4-pdec-bad-window-classification-lemma.md`.

It combines the unified PDEC-or-SAE dichotomy, column-conflict routing, tail-anchor persistence, and Rankin access rules to split each named low-mod bad-window family into SAE, LHB-PDEC, or Routed-PDEC. This is still a routing theorem: the non-LHB exits and the final dual $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ certificate remain open. The homogeneous-splitting rule is recorded in

`docs/monograph/h4-pdec-homogeneous-splitting-lemma.md`.

It closes the bookkeeping case in which one attempted certificate mixes different (Q, τ, F, κ, p) types: such a family must be partitioned into homogeneous subfamilies before applying UPS-1 or the bad-window classification. Thus mixed type is not an additional mathematical exit; the remaining exits are SAE, ColumnCRT/ColumnRadius, TailAnchor, and Rankin. The column-defect routing contract is recorded in

`docs/monograph/h4-pdec-column-defect-routing-contract.md`.

It defines the conditional certificate objects for ColumnRadius and ColumnCRT rows: radius rows use a threshold D_0, a column-witness selector, and phase-compatible weights $W_D(t)$, while displacement rows use a label ℓ, a non-zero displacement residue a, a threshold L_D, and phase-compatible weights $W_{\ell,a}(t)$. Violating such a row routes to the named ColumnRadius or ColumnCRT exit after the zero displacement class has been separated by the same-column prime-witness rigidity. This is a routing-contract closure only; the exits themselves still require materialized weights, thresholds, and exclusion certificates. The first finite materialization of those weights is recorded in

`docs/monograph/h4-pdec-column-defect-weight-certificate.json`.

It uses the refined finite phase $\tau_{\mathrm{fin}} = (p, q, \mathrm{row})$ for $p \leq 1000$ and the five tightest RCI rows for each p. In this finite domain, the blocks $D_{\mathrm{col}} > 81$ and displacement-residue load > 2 are both empty, over 835 phases. This is a finite refined-phase certificate; it does not provide a global D_0, L_D theorem or exclude the ColumnRadius/ColumnCRT exits. The recursive zero-row peeling route is isolated in

`docs/monograph/prime-matrix-recursive-peeling-zero-row-hardpoint.md`.

It proves the exact layer-peeling identity: after a p-sieve zero window is peeled to the previous prime level, any resurrected survivor must be a multiple of the peeled prime with rough cofactor. Thus the recursive object is a punctured zero window, not automatically a lower-level zero row. The accompanying finite audit finds no forced consecutive lower zero rows in the known first-zero-row samples. The useful remaining target is therefore an absorption theorem routing those resurrected points to ColumnCRT, ColumnRadius, or TailAnchor, rather than a direct short-period contradiction. The adjacent-shell refinement is recorded in

`docs/monograph/prime-matrix-adjacent-shell-recursive-descent-route.md`.

For adjacent primes $p < q$, it proves that the only composite old p-sieve survivor in $[1, q^2]$ is q^2. Hence a non-first q-zero row first descends to an old p-sieve q-window, with only the terminal q^2 puncture in the last row. That window either contains a full p-aligned zero row or is a seam window whose two guard intervals have lengths a and $p - (q - p) - a$. The resulting narrow target is SeamGuard-Elimination: guards cannot absorb all forced survivors without entering SAE, PDEC, or ColumnCRT. The multi-level seam descent audit is recorded in

`docs/monograph/prime-matrix-seam-multilevel-descent-route.md`.

In this conditional model, if a seam interval is already an old p-sieve zero window, then after descending to a lower prime $h < p$ the only resurrected points are those with $P^-(n) \in (h, p]$, together with the terminal q^2 puncture in the last q-row. Any full h-aligned row avoiding these resurrected points is therefore a forced h-zero row. The finite ledger checks all 21339 seam windows for $p \le 500$ and a deterministic $p \le 2000$ sample of 11488 seam windows, with no persistent blocking cases. This is not a global theorem; it sharpens the remaining gate to the SMD-Global Inequality, or else a proof that persistent revival blocking enters SAE, PDEC, or ColumnCRT. The tail-mirror refinement is recorded in

`docs/monograph/prime-matrix-seam-tail-mirror-descent-route.md`.

For a forced lower h-zero row with row number R, set $N_h = M_h/h$ and $\rho = ((R - 1) \bmod N_h) + 1$. If $\rho \le h$ or $N_h - \rho + 1 \le h$, then periodicity or CRT reflection places a conditional zero row inside the lower $h \times h$ matrix. The finite ledgers above again show full success in the same domains. This motivates the sharper TailMirror-SMD gate: every genuine seam counterexample must hit such a lower head/tail phase, or its persistent non-hit phases must be routed to SAE, PDEC, or ColumnCRT. The all-branch version is recorded in

`docs/monograph/prime-matrix-total-zero-row-recursive-descent-route.md`.

It treats aligned containment, seam windows, and the terminal q^2 puncture by the same descent rule. In the $p \le 500$ ledger all 21936 non-first q-rows, including 597 aligned rows and 21339 seam rows, force a lower matrix head/tail phase. In the deterministic $p \le 2000$ sample all 11583 checked rows do the same. The remaining theorem-level statement is TotalDescent-TM: every genuine higher zero row must force a lower $h \times h$ zero row by periodicity or tail reflection, or else its non-hit phase set must enter SAE, PDEC, or ColumnCRT. The fixed $h = 3$ refinement is recorded in

`docs/monograph/prime-matrix-total-descent-h3-margin-route.md`.

Each complete 3-row contains a unique candidate c_R coprime to 6. If $5 \le P^-(c_R) \le p$, that row is blocked by a resurrected point when descending from the p-sieve to the 3-sieve; if $P^-(c_R) > p$, then

c_R is already a survivor of the old p-sieve and contradicts the assumed p-sieve zero window. Thus the narrowest remaining form of TotalDescent is the six-wheel inequality

$$\#\{R : J_R \subset I_s\} > \#\{R : J_R \subset I_s,\ 5 \le P^-(c_R) \le p\}.$$

The audit

`docs/monograph/prime-matrix-total-descent-h3-margin-audit.md`

checks this for all 1,552,462 non-first q-rows with $p \le 5000$, with minimum margin 1. This is a finite certificate and a sharpened interface, not a global theorem: a final proof must establish the six-wheel inequality uniformly, or route full blocking to SAE/PDEC/ColumnCRT. The hard-attack reduction

`docs/monograph/prime-matrix-h3-sixwheel-hard-attack.md`

clarifies the remaining obstruction. The direct six-wheel inequality sits at the Buchstab/linear-sieve boundary $u = 2$, i.e. at square-root interval length near $x = q^2$. Therefore a proof cannot be obtained by a generic short-interval theorem or a finite template. If the six-wheel candidates are all blocked, they admit the first-factor partition $n = \ell m$ with $5 \le \ell \le p$, m in an interval of length q/ℓ, and $P^-(m) \ge \ell$. The next theorem-level gate is consequently an H3 full-blocking defect theorem: high first-factor load enters Tail/PDEC, distributed low-load coverage produces low-mod PDEC energy, and endpoint residue concentration is SAE or ColumnCRT. The near-miss audit

`docs/monograph/prime-matrix-h3-full-blocking-defect-audit.md`

checks the same ledger for all $p \le 5000$. Among 1,552,462 non-first q-rows, only 4015 have margin at most 20, and these occur only up to $p = 317$ and $q = 331$. Their aggregate first-factor load is led by the small-prime skeleton $5, 7, 11, 13, \ldots$. This data does not prove the theorem, but it identifies the correct next split: small-first-factor overload should be absorbed by Tail/PDEC, while low-load many-label blocking should produce low-mod PDEC/ColumnCRT energy. The small-factor envelope route

`docs/monograph/prime-matrix-h3-small-factor-envelope-route.md`

makes this split deterministic. For a cutoff y, let C_y be the load carried by first factors at most y, let $R_y = A - C_y$, and let $\ell_+(y)$ be the next prime. If full blocking holds, then either C_y is already a small-skeleton overload, or at least

$$K_y = \left\lceil \frac{R_y}{\lfloor q/\ell_+(y) \rfloor + 1} \right\rceil$$

middle-tail labels are active. The finite ledger for the near-miss rows gives maximum forced values $K_y = 7$ for $y = 31$ and $K_y = 9$ for $y = 43$. The remaining proof obligation is to convert small-skeleton overload into Tail/PDEC, or many active labels into H3-PDEC energy, with endpoint losses routed to SAE/ColumnCRT. The global scaling audit

`docs/monograph/prime-matrix-h3-global-scaling-law-route.md`

records the underlying scale. The natural six-wheel rough count is

$$\#A_s \prod_{5 \le \ell \le p} \left(1 - \frac{1}{\ell}\right) \asymp \frac{q}{\log q}.$$

In the finite ledger, from $p = 331$ onward every prime layer has minimum H3 margin at least 21, and bucket averages track this Mertens scale. Thus a formal counterexample must annihilate a growing $q/\log q$ rough-residue scale. The current proof task is to show that such annihilation forces small-skeleton Tail/PDEC overload, many-label H3-PDEC energy, or endpoint SAE/ColumnCRT. The constant-level formula is recorded in

`docs/monograph/prime-matrix-h3-universal-scaling-inequality.md`.

In the finite ledger, $M_{H3}(p, s) \geq 0.30\, q/\log q$ holds from $p = 113$ onward, and $0.40\, q/\log q$ holds from $p = 2011$ onward. These are not asserted as unconditional short-interval prime theorems. They are target constants for a defect inequality: if the H3 margin falls below $cq/\log q$, then SmallSkeletonOverload, ManyLabel-PDEC, or Endpoint-SAE/ColumnCRT must fire. The global tail-energy lemma

`docs/monograph/prime-matrix-h3-global-tail-energy-lemma.md`

proves the deterministic core of this defect inequality. For any cutoff y, target B, threshold L, and modulus

$$d \geq 2\left(\left\lfloor \frac{q}{\ell_+(y)} \right\rfloor + 1\right),$$

where $\ell_+(y)$ is the next prime after y, the implication

$$M_{H3}(p, s) < B \quad \Longrightarrow \quad C_y > \#A_s - B - 2L \quad \text{or} \quad E_y(d) > L$$

holds for every adjacent pair $p < q$ and every row window $2 \leq s \leq q$. Substituting $B = cq/\log q$ and $L = \lambda q/\log q$ gives a uniform infinite-scale defect formula. This is still not the final H3 lower bound: the remaining theorem-level exits are SmallSkeletonOverload $\Rightarrow$ Tail/PDEC and TailEnergy $\Rightarrow$ H3-PDEC/ColumnCRT. The first exit is bridged in

`docs/monograph/prime-matrix-h3-small-skeleton-pdec-bridge.md`.

Let

$$V_y = \#A_s \prod_{5 \leq r \leq y} \left(1 - \frac{1}{r}\right), \qquad D_y = V_y - R_y,$$

where $R_y = \#\{n \in A_s : P^-(n) > y\}$. The overload inequality $C_y > \#A_s - B - 2L$ gives $R_y < B + 2L$, hence

$$D_y > V_y - B - 2L.$$

Thus, whenever $V_y \geq (c + 2\lambda + \eta)q/\log q$, small-skeleton overload forces a PDEC defect of size $D_y > \eta q/\log q$. The remaining issue is not this bridge, but the exclusion of such persistent PDEC defects and of the tail-energy alternative. The tail alternative is put in Fourier form in

`docs/monograph/prime-matrix-h3-tail-energy-fourier-bridge.md`,

where Parseval gives $\mathcal{F}_y(d) = dE_y(d)$ for the non-zero residue frequencies of the tail first-factor labels. The synthesis

`docs/monograph/prime-matrix-h3-unified-defect-criterion.md`

therefore proves the conditional closure criterion

$$D_y \leq V_y - B - 2L \quad \text{and} \quad \mathcal{F}_y(d) \leq dL \quad \Longrightarrow \quad M_{H3}(p, s) \geq B.$$

With $B = cq/\log q$ and $L = \lambda q/\log q$, this is the current global scale theorem: the empirical $q/\log q$ law is explained by two explicit low-mod defect exclusions. The unconditional row theorem still requires proving those exclusions uniformly. The source of the $q/\log q$ scale is clarified in

`docs/monograph/prime-matrix-h3-first-row-scale-bridge.md`.

Let $\Pi_1(q) = \pi(q)$ be the number of primes in the first row up to q. By the prime number theorem, $\Pi_1(q) \sim q/\log q$. On the other hand, adjacent-shell singleton rigidity says that below q^2 every old p-sieve survivor, apart from the endpoint q^2, is prime. Hence the average H3 margin over the non-first q-rows satisfies

$$\frac{1}{q-1}\sum_{s=2}^{q} M_{H3}(p,s) \sim \frac{q}{2\log q} \sim \frac{1}{2}\Pi_1(q).$$

The factor $1/2$ is simply $\log q/\log(q^2)$. Thus H3 is the square-shell projection of the first-row prime scale; a zero H3 row would have to remove a full first-row-scale mass from one row and therefore must be accounted for by the low-mod defects above. The pointwise boundary is recorded in

`docs/monograph/prime-matrix-h3-pointwise-closure-boundary.md`.

The average identity above is rigorous, but it does not imply pointwise positivity. The missing statement is a pointwise scale-transfer theorem, for example

$$M_{H3}(p,s) \geq \kappa\,\pi(q) \qquad (2 \leq s \leq q)$$

for some fixed $\kappa > 0$, or equivalently the uniform exclusion of the D_y and $\mathcal{F}_y(d)$ defects in the H3 unified criterion. The analytic boundary is recorded in

`docs/monograph/prime-matrix-h3-square-root-short-interval-barrier.md`.

It identifies this pointwise transfer with a prime lower bound in aligned intervals of length $q = \sqrt{q^2}$ near $x = q^2$. This is exactly the square-root short-interval scale, where ordinary average PNT input and the linear sieve at $u = 2$ do not supply a pointwise positive lower bound. The remaining theorem is therefore a genuine square-root defect exclusion, not a bookkeeping consequence of the first-row average. The direct hard attack is archived in

`docs/monograph/prime-matrix-h3-square-root-defect-exclusion-hard-attack.md`.

It shows that the D_y branch can be controlled at a small sieve level by the standard one-dimensional interval-sieve fundamental lemma. The obstruction is the tail branch: for a natural low modulus d, the tail Fourier energy satisfies

$$\frac{1}{2}T_y \leq E_y(d) \leq T_y,$$

so it is essentially the tail-label count. The final hard core is therefore to exclude tail labels and double-rough points as an exact full-row filler, or to force them into ColumnCRT, endpoint phase, or cofactor-structure defects. The local rigidity of such a filler is recorded in

`docs/monograph/prime-matrix-h3-tail-filler-rigidity-hardcore.md`.

Adjacent H3 candidates differ by 2 or 4, and candidates two steps apart differ by 6. If both endpoints are tail-filled, all their large factor families are disjoint. A repeated tail label must be separated by at least $y/4$ candidate steps, and any block of diameter less than y consumes at least two fresh large prime factors per candidate. Thus the remaining obstruction is the global concatenation of these local 2/4/6 rough short-difference units into a whole row. The global concatenation interface is sharpened in

`docs/monograph/prime-matrix-h3-tail-filler-global-chain-capacity.md`.

For any continuous tail-filled block B, full concatenation forces

$$\mathrm{EdgeCap}(B;y)=|B|-1, \qquad \mathrm{TriCap}(B;y)=|B|-2.$$

Here EdgeCap counts compatible adjacent 2/4 tail edges and TriCap counts compatible 2/4/6 triples. If $y>\sqrt{q+6}$, a fixed ordered tail-label pair contributes at most once in the row; if $y>q^{2/3}$, every tail-filled point is a double-rough semiprime. Hence the final H3 hard input can now be stated as an explicit capacity exclusion: prove that these edge or triple capacities are strictly below the full-chain requirements, or that full capacity forces PDEC, ColumnCRT, endpoint phase, or cofactor-structure defects. This is a necessary-capacity reduction, not yet an unconditional proof of H3. The macroscopic full-tail branch is then attacked in

`docs/monograph/prime-matrix-h3-tail-edge-selberg-exclusion.md`.

For a fixed even shift $\delta\in\{2,4,6\}$ and an interval I of length H, the classical two-dimensional Selberg upper-bound sieve gives

$$\#\{n\in I:(n(n+\delta),P(y))=1\}\le A_\delta(\theta)\frac{H}{(\log y)^2} \qquad (3\le y\le H^\theta).$$

Every filled adjacent edge gives a y-rough pair with shift 2 or 4, and every filled triple gives a y-rough endpoint pair with shift 6. Hence a continuous tail-filled block B must satisfy

$$|B|\le 2+A_*(\theta)\frac{q+6}{(\log y)^2}.$$

Thus a macroscopic block of length comparable with q, in particular a whole H3 row of about $q/3$ candidates, cannot be filled only by tail labels at sufficiently large scale. The remaining H3 obstruction is consequently narrower: any surviving bad row must split tail points into many short blocks, and the small-factor cuts or short-block endpoint phases must be routed to PDEC, ColumnCRT, endpoint, or cofactor defects. The short-block routing is made exact in

`docs/monograph/prime-matrix-h3-shortblock-singleton-barrier.md`.

If the tail points are decomposed into maximal consecutive tail blocks, with total tail mass T, number of blocks J, and internal adjacent-tail edges E, then

$$T=J+E.$$

The two-dimensional sieve controls E, so a bad row with T still of order $q/\log y$ would have almost all tail mass in singleton tail blocks. The small-factor cuts themselves are not yet a PDEC contradiction, because the small skeleton has natural size comparable with q. The final local obstruction is therefore the singleton-tail system: a point $a_j=\ell_j m_j$ with $\ell_j>y$ is isolated between two small-skeleton neighbours, so $a_j\equiv 0 \pmod{\ell_j}$ and simultaneously satisfies two non-zero congruences modulo small labels $r_-,r_+\le y$. One must prove that many such squeezed singleton phases force PDEC, ColumnCRT, endpoint, or cofactor defects. The squeezed singleton phase is converted into a CRT capacity criterion in

`docs/monograph/prime-matrix-h3-singleton-clamp-defect-criterion.md`.

For a fixed clamp datum $c = (\delta_-, \delta_+, r_-, r_+)$, the two small-skeleton congruences are either incompatible or define one class $\rho(c) \pmod{\operatorname{lcm}(r_-, r_+)}$. Adding the tail label $\ell > y$ gives one class modulo $\ell \operatorname{lcm}(r_-, r_+)$, hence one row contains at most

$$1 + \left\lfloor \frac{q + O(1)}{\ell \operatorname{lcm}(r_-, r_+)} \right\rfloor$$

points from that unit. Consequently many singleton tails must enter one of three branches: tail-label concentration, clamp low-mod concentration, or distributed singleton capacity. The first two are named defect entrances. The final unresolved branch is the distributed signed excess of double-rough semiprimes over the prime survivor channel. This distributed branch is made bilinear in

`docs/monograph/prime-matrix-h3-distributed-singleton-bilinear-obstruction.md`.

For $y > q^{2/3}$, every singleton tail is a product ℓm with $y < \ell \le p$ and m prime. The clamp class $\rho(c) \pmod{R(c)}$ is equivalent to the moving cofactor congruence

$$m \equiv \ell^{-1}\rho(c) \pmod{R(c)}.$$

Thus the final distributed obstruction is a signed bilinear prime-counting defect over the short cofactor intervals I/ℓ. Without tail-label concentration, clamp concentration, endpoint accumulation, or ColumnCRT/cofactor bias, one must prove that this signed bilinear error is only $O(q/\log^2 y)$; this is the current last H3 input. The non-zero-frequency bridge is recorded in

`docs/monograph/prime-matrix-h3-bilinear-large-sieve-defect-bridge.md`.

Expanding the cofactor residue condition modulo $R(c)$ into non-principal characters gives

$$\chi(\ell^{-1}\rho(c)) = \chi(\rho(c))\overline{\chi(\ell)}.$$

Hence a large distributed singleton error forces a non-principal bilinear character sum over the short cofactor intervals. A plain large-sieve bound is still too weak at this short window scale; the final input is a dispersion/Kloosterman-type estimate for this family, or a proof that the corresponding non-zero frequency is already a PDEC, ColumnCRT, or cofactor defect. The additive Kloosterman reduction is

`docs/monograph/prime-matrix-h3-dsb-kloosterman-window-reduction.md`.

The congruence $m \equiv \rho(c)\bar{\ell} \pmod{R(c)}$ gives, after finite additive Fourier expansion, the inverse phase

$$e\left(-\frac{h\rho(c)\bar{\ell}}{R(c)}\right).$$

Thus the final H3 frequency input is a short-window Kloosterman sum with variable ℓ, cofactor window I/ℓ, modulus $R(c)$, and frequency h. The remaining audit splits into four branches: KLS-window coverage of the active parameters, high-lcm clamp escape, high-frequency endpoint tail, and coefficient concentration. The high-lcm branch is routed further in

`docs/monograph/prime-matrix-h3-dsb-high-lcm-clamp-routing.md`.

If $R(c) > R_0$, then one row contributes at most

$$1 + \left\lfloor \frac{q + O(1)}{R_0} \right\rfloor$$

points from a fixed clamp unit. Hence any high-lcm mass of order $q/\log y$ must activate many almost-singleton high-lcm units. Across a persistent family of bad rows this gives a non-zero CRT/Fourier defect, hence a PDEC/ColumnCRT exit; if it is not persistent, it is a sparse single-window escape. This does not yet exclude the branch, but it removes high-lcm mass from the low-modulus KLS-window estimate and routes it to the existing final exits. The common energy source of the two high-lcm exits is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-fourier-energy-clamp.md`.

For a homogeneous high-lcm modulus block, with residue-counting measure μ and total mass U, finite Plancherel gives

$$\sum_{1 \le h < R} |\widehat{\mu}(h)|^2 = R \sum_{a \bmod R} \mu(a)^2 - U^2.$$

In particular, when $U \le R/2$, this non-zero energy is at least $RU/2$. Hence high-lcm escape necessarily injects Fourier energy: persistent energy must be excluded by PDEC/ColumnCRT, and isolated energy by SAE/endpoint analysis. The persistent branch is converted to a PDEC threshold in

`docs/monograph/prime-matrix-h3-dsb-hlc-pdec-threshold-bridge.md`.

For one homogeneous formal unit B with phase-counting vector g_B,

$$L_{\mathrm{HLC}}(B) = \left(\frac{R \sum_a g_B(a)^2 - U_B^2}{R - 1} \right)^{1/2}$$

is a forced non-zero-frequency lower bound. Thus a persistent high-lcm unit is excluded by a same-unit certificate $U_{\mathrm{CRT}}(B) < L_{\mathrm{HLC}}(B)$. The case $U_B > R/2$ is not a new escape; it is dense effective-modulus concentration and routes back to KLS-window, PDEC/ColumnCRT, or SAE/endpoint. If this PDEC upper certificate fails, the failure is localized in

`docs/monograph/prime-matrix-h3-dsb-hlc-pdec-failure-localization.md`.

For a failing non-zero frequency h and direction ζ, every cap

$$\mathcal{C}_\alpha = \{a : \Re(\zeta e(ha/R)) \ge \alpha\}, \qquad 0 \le \alpha < L/U,$$

has mass at least $(L - \alpha U)/(1 - \alpha)$. Hence failure is a concrete Bohr-cap concentration, which must become a same-unit PDEC constraint or a sparse SAE/endpoint certificate. The cap is further decomposed in

`docs/monograph/prime-matrix-h3-dsb-hlc-bohr-cap-component-route.md`.

With $d = (h, R)$ and $R' = R/d$, the cap is the d-fold lift of one arc in $\mathbb{Z}/R'\mathbb{Z}$. Each lifted component has length at most $d + \omega(\alpha)R$, where $\omega(\alpha) = \arccos(\alpha)/\pi$, and one component carries mass at least $(L - \alpha U)/(d(1 - \alpha))$. Thus the remaining cap obstruction is either a large-gcd low-effective-modulus concentration or a short-arc component concentration. The large-gcd branch is descended in

`docs/monograph/prime-matrix-h3-dsb-hlc-high-gcd-descent.md`.

If $d = (h, R) > 1$ and $R' = R/d$, the projected vector $g'(b) = \sum_{a \equiv b \pmod{R'}} g(a)$ satisfies

$$\widehat{g}_R(h) = \widehat{g}_{R'}(h/d),$$

and the Bohr-cap mass is preserved. Hence each high-gcd step strictly lowers the effective modulus and must terminate in a low-effective-modulus PDEC/KLS exit or in the short-arc case. The short-arc case is quantified in

`docs/monograph/prime-matrix-h3-dsb-hlc-short-arc-density-pressure.md`.

If a short arc has length at most $D_0 + \omega(\alpha)R$ and mass forced by the Bohr-cap lower bound, then its density relative to the global density U/R is at least

$$\frac{R(L - \alpha U)}{D_0(1-\alpha)U(D_0 + \omega(\alpha)R)}.$$

If this pressure exceeds $1 + \epsilon$, a local-density PDEC row or SAE/endpoint certificate is forced; otherwise only the explicit parameter residual remains. The low-pressure residual is optimized in

`docs/monograph/prime-matrix-h3-dsb-hlc-short-arc-pressure-optimizer.md`.

Writing $t = L/U$ and

$$\Psi_{D_0,R}(t) = \sup_{0 \le \alpha < t} \frac{t - \alpha}{D_0(1-\alpha)(D_0/R + \omega(\alpha))},$$

the residual is the one-dimensional condition $\Psi_{D_0,R}(t) \le 1 + \epsilon$. This implies the same-unit flatness estimate

$$\sum_a g(a)^2 \le \frac{1 + (R-1)\tau^2}{R} U^2$$

and the corresponding support lower bound. Thus low-pressure short-arc mass is routed back to the KLS-window coefficient-norm check or to coefficient concentration. The KLS admission step is separated in

`docs/monograph/prime-matrix-h3-dsb-hlc-l2-flat-kls-admission.md`.

It lists six required checks for an HLC L2-flat residual: effective modulus, effective frequency, endpoint smoothing, coefficient L^2 norm, gcd/unit strata, and dyadic or tail-label splitting. The L2-flat estimate supplies the coefficient-norm check; any other failed item is a named exit, and only if all six pass may the KLS-window input be invoked. Clean admission is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-clean-kls-reduction.md`.

A clean HLC formal unit is one in which the six named exits do not occur; then the K1–K6 admission checks pass. The remaining object is the clean HLC Kloosterman window

$$\mathcal{K}_{\mathrm{HLC}}(B),$$

and the remaining external/self-contained input is an estimate $\mathcal{K}_{\mathrm{HLC}}(B) = O(q/\log^2 y)$, denoted HLC–KLS-ext in the notes. The external-theorem adaptation is now separated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-external-adaptation.md`.

It matches the clean HLC object to a DI/BFI/Kuznetsov-type windowed Kloosterman input: the inverse phase is $e(-h\rho(c)\overline{\ell}/R(c))$, the modulus and frequency checks are supplied by K1–K2, endpoint smoothing by K3, coefficient L^2 control by K4, gcd/unit strata by K5, and dyadic or tail-label losses by K6. Thus the clean HLC branch is closed in the external-deep-theorem version. A fully self-contained version would still require reproving the corresponding spectral/dispersion Kloosterman theorem inside this manuscript. The self-contained obligation is further compressed in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-core-reduction.md`.

There the clean HLC window is decomposed into dyadic Type-I/II blocks with standard kernel $\mathfrak{S}(\mathcal{D})$, and the proof of HLC–KLS-ext is reduced to one core estimate $|\mathfrak{S}(\mathcal{D})| \leq \mathcal{N}(\mathcal{D})/\log^A y$. Thus the remaining no-black-box line is this single windowed Kloosterman spectral mean estimate. The no-black-box spine is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-core-self-contained-spine.md`.

It completes the smooth completion, Kloosterman quadratic-form reduction, coefficient L^2 bookkeeping, and the implication from the Kuznetsov large-sieve atom to the core estimate. It also records why pointwise Weil bounds alone do not yield the required arbitrary logarithmic saving. Hence a fully self-contained version now has one precise remaining analytic line: prove the specialized Kuznetsov large-sieve atom for this window family. That line is expanded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kuznetsov-ls-atom-expansion.md`.

The atom is decomposed into smoothing, the specialized Kuznetsov trace formula, Bessel transform decay, the spectral large sieve including oldforms and Eisenstein spectrum, and the well-factorable dispersion step that supplies the logarithmic saving. The manuscript therefore records the exact no-black-box obligations rather than hiding them under the single phrase "spectral theory". The Bessel-transform part is now treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-c-bessel-transform-decay.md`.

It uses the Bessel differential equation, the self-adjoint Bessel operator, and repeated integration by parts to obtain the required rapid transform decay. The remaining no-black-box analytic atoms at this point are the specialized Kuznetsov formula, the spectral large sieve, and the well-factorable dispersion logarithmic saving. The spectral-large-sieve atom is further reduced in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-d-spectral-large-sieve-spine.md`.

This dualizes the spectral large sieve to a spectral projection kernel, accounts for oldforms, Eisenstein spectrum and holomorphic forms as polylogarithmic losses, and reduces KZ–D to a single pre-trace kernel bound with diagonal size T^2 and off-diagonal Schur size N_0. The pre-trace kernel is further decomposed in

`docs/monograph/prime-matrix-h3-dsb-hlc-ptk-d-pretrace-kernel-bound.md`.

There the diagonal term is recorded as the local Weyl volume, and the remaining off-diagonal part is reduced to a spatial lattice-point/correlation row-column bound, denoted LPC–D. The spatial kernel is split in

`docs/monograph/prime-matrix-h3-dsb-hlc-lpc-d-spatial-correlation-split.md`.

The far tail is absorbed by kernel decay, the identity-near sector is empty or parabolic, and the parabolic cusp sector is controlled by divisor bookkeeping. The remaining spatial line is the generic hyperbolic local-correlation bound GHLC–D. That last spatial line is now treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-ghlc-d-local-schur-closure.md`.

The key correction is to estimate the pre-trace object as an integral kernel, not by pointwise hyperbolic neighbour counting. On the local radius $r_* = T^{-1} \log^B y$ one has

$$T^2 \int_0^{r_*} (1 + Tr)^{-A} \sinh r \, dr = O_A(1),$$

so the local ball area cancels the T^2 height of the kernel. Combining this local L^1 kernel mass with the Poincare-packet Schur normalization gives GHLC–D, hence LPC–D, PTK–D, and KZ–D. At that point the remaining analytic atoms were KZ–B, the specialized Kuznetsov trace formula, and KZ–E, the well-factorable dispersion logarithmic saving. The KZ–B specialization is treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-b-kuznetsov-trace-specialization.md`.

It derives the needed trace formula from the automorphic kernel, Poincare-packet unfolding, the double-coset Kloosterman geometric side, spectral Plancherel, and Bessel-transform normalization. This closes the formula-conversion part; it does not supply the final logarithmic saving. Hence the remaining no-black-box analytic atom in the HLC branch is KZ–E, the well-factorable dispersion logarithmic-saving step. The current KZ–E attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-e-well-factorable-dispersion-spine.md`.

It internalizes the well-factorable convolution algebra, the dispersion variance identity, the CRT reduction to a standard Kloosterman phase, and the gcd/endpoint polylogarithmic ledger. This proves that KZ–E would follow from one remaining estimate, denoted WFD-core: a windowed well-factorable Kloosterman dispersion mean estimate. That core is further balanced in

`docs/monograph/prime-matrix-h3-dsb-hlc-wfd-core-balanced-factor-reduction.md`.

Using the square-root well-factorable decomposition $\lambda_c = \sum_{c=uv} \alpha_u \delta_v$, the condition $c \asymp C$ forces $u, v \asymp C^{1/2}$ up to polylogarithmic boundary blocks. The gcd strata are polylogarithmic, and CRT factors the phase modulo uv into coupled phases modulo u and v. At this intermediate stage the remaining atom is BWFD-core, a balanced two-modulus well-factorable Kloosterman dispersion mean estimate. The subsequent completion attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bwfd-core-spectral-completion-attack.md`.

It performs exact finite Fourier completion in the s variable and uses complete Kloosterman multiplicativity modulo uv. This proves that BWFD-core follows from one still narrower atom, BSC-core: balanced complete Kloosterman bilinear correlation with arbitrary logarithmic saving. Ordinary KZ–D spectral large sieve recovers only the raw quadratic norm scale, and pointwise Weil bounds give only single-sum square-root cancellation; neither supplies the required $\log^{-A}$ saving. The current direct attack on BSC-core is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bsc-core-kloosterman-fraction-attack.md`.

It expands the complete Kloosterman sums term by term and exposes the reciprocal-fraction phase $e(\bar{v}R/u+\bar{u}T/v)$. One-sided degeneracy is localized to quadratic congruences such as $(a_h+\ell)x^2+b_h \equiv 0 \pmod{u}$; the remaining no-black-box atom is KFLS-core, a balanced Kloosterman-fraction large-sieve estimate with arbitrary logarithmic saving. The next square-kernel attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kfls-core-square-kernel-attack.md`.

It expands the square of the reciprocal-fraction sum into a four-modulus phase kernel. This also rules out an absolute Schur proof of the full kernel: the exact diagonal recovers only the raw quadratic norm scale. The remaining atom is therefore CFQK-core, a centered four-modulus Kloosterman-fraction correlation estimate covering the semi-diagonal and true off-diagonal layers. The block-centering correction is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-cfqk-core-block-centering-attack.md`.

It shows that the entire same-(u,v) block diagonal is local variance and cannot be forced to be logarithmically small for arbitrary admissible coefficients. The remaining obligations are therefore BD-CEN, the block-centering identity, OSQK-core for one shared modulus, and TFQK-core for the true four-modulus layer. The CRT lift-parity variant is audited in

`docs/monograph/prime-matrix-crt-lift-parity-audit.md`.

For adjacent primes $p < q$, the non-trivial q-columns are not covered by the new prime q; therefore a hypothetical q-row zero does descend to an old p-sieve zero window of length q. However, the old p-sieve row period already has an even number of zero rows by reflection symmetry, so multiplying the period by the odd factor q preserves evenness and does not create a parity contradiction. The retained conclusion is only the early-window dichotomy: an aligned lower zero row or a seam window routed to PDEC/SAE. The attempted recursive continuation is audited in

`docs/monograph/prime-matrix-recursive-mirror-descent-audit.md`.

It separates three objects: zero rows in a full CRT period at fixed width q, zero rows after peeling the largest base prime, and zero rows in an actual smaller-width matrix. These objects do not imply one another. In the finite audit, full-period zero rows occur for several adjacent pairs, but no early q^2 zero row and no actual smaller matrix zero row is forced. The next usable gate remains seam-window exclusion, not recursive parity. The annulus-mirror localization audit

`docs/monograph/prime-matrix-annulus-mirror-localization-audit.md`

separates two reflections. The CRT reflection preserving zero residue classes is $n \mapsto -n \pmod{M_p}$ and sends a small row to the tail of the CRT period. The terminal reflection $n \mapsto q^2 - n$ does land in an early interval, but it changes the forbidden classes from 0 to $q^2 \bmod \ell$. It is therefore a terminal non-zero-class block, not a smaller prime-matrix zero row. The scaled-row and half-width version is audited in

`docs/monograph/prime-matrix-scaled-peeling-halfwidth-audit.md`.

It shows that the scaled height nP/p is only an approximate locator: a lower aligned zero row appears only when the upper window fully contains a lower row. In the same samples, passing to a prime near half the upper width gives no automatic pair of consecutive lower zero rows, because the half-level sieve resurrects points whose least factor lies above the half-width level. The resurrected-point absorption route is now recorded in

`docs/monograph/prime-matrix-rpz-absorption-defect-route.md`

with the finite audit

`docs/monograph/prime-matrix-rpz-absorption-defect-audit.md`.

It routes complete absorption into five alternatives: a survivor remains, TailAnchor load, ColumnCRT displacement load, ColumnRadius, or a residual Distributed-RPZ branch. In the known first-zero-row samples the half-level has 15 resurrected points, no missing absorber, and maximum one-window label and top-column loads equal to 1. Thus the route does not justify a single-window TailAnchor exclusion; the next theorem-level gate is the Distributed-RPZ capacity barrier, or a proof that distributed absorption necessarily returns to one of the named column or tail exits. The first attack on this gate is recorded in

`docs/monograph/prime-matrix-rpz-sliding-plateau-barrier.md`.

It proves a sliding-platform multiplicity formula: if a zero window remains zero under a consecutive block of shifts, then a resurrected source in the common core repeats with the full platform multiplicity. Hence distributed absorption on such a platform is either a persistent TailAnchor load or, after source deletion, a boundary-compressed branch. The audit in

`docs/monograph/prime-matrix-rpz-sliding-plateau-audit.md`

finds platform length 5 in the known samples and TailAnchor triggering at $T_0 = 4$ in all five cases. The remaining gate is therefore the boundary-compressed RPZ branch, not unrestricted distributed absorption. The boundary-compressed branch is further routed in

`docs/monograph/prime-matrix-rpz-boundary-compressed-core-route.md`.

It proves that, if TailAnchor is excluded, the center interval $J_{T_0} = [L + A + T_0, R + B - T_0]$ is an h-sieve zero interval. The audit

`docs/monograph/prime-matrix-rpz-bcb-core-audit.md`

finds that all 11 center survivors in the known samples are TailAnchor core sources; after conditional TailAnchor deletion, all five center intervals are clean and contain an aligned half-width row. The next gate is therefore the BCB grid/endpoint exclusion: either the center contains an aligned lower zero row, or the endpoint seam enters SAE, PDEC, or ColumnCRT. The exact grid criterion is recorded in

`docs/monograph/prime-matrix-rpz-bcb-grid-endpoint-criterion.md`.

For $J = [u, v]$, $N = v - u + 1$, and $\delta_h(u) \equiv 1 - u \pmod h$, the interval contains an aligned h-row if and only if $N \geq \delta_h(u) + h$. Failure is therefore a finite endpoint seam phase. The audit in

`docs/monograph/prime-matrix-rpz-bcb-grid-endpoint-audit.md`

matches this criterion in all five samples and finds no endpoint-defect sample. The remaining gate is endpoint persistence exclusion: a failed grid phase must be isolated as SAE or persistent as PDEC/ColumnCRT. This endpoint gate is routed in

`docs/monograph/prime-matrix-rpz-bcb-endpoint-persistence-route.md` and audited in

`docs/monograph/prime-matrix-rpz-bcb-endpoint-phase-ledger.md`.

For fixed (h, N), endpoint failure is a finite residue ledger in $u \bmod h$. In the known samples there are no actual endpoint failures and only two possible failure phases. Sparse loads are SAE obligations; persistent same-phase loads are PDEC/ColumnCRT objects. Thus the RPZ endpoint branch is now named, not closed: the remaining work is either certificate closure for SAE/PDEC/ColumnCRT or recursive descent from the lower h-zero row. The two remaining RPZ attacks are now advanced together in

`docs/monograph/prime-matrix-rpz-dual-track-closure-route.md`.

Track A fixes the certificate interfaces for RPZ-SAE, RPZ-PDEC, and RPZ-ColumnCRT. Track B uses the lower-zero descent audit

`docs/monograph/prime-matrix-rpz-lower-zero-descent-audit.md`

which shows that, in the known BCB samples, all six conditional lower zero rows descend to a $p = 2$ impossible zero row with no endpoint/grid obstruction. This is still a sample-level descent audit; the global task is to prove descent-grid persistence or route every obstruction to the named certificate interfaces. The obstruction side is now made explicit in

`docs/monograph/prime-matrix-rpz-lower-descent-obstruction-ledger.md`

and

`docs/monograph/prime-matrix-rpz-certificate-materialization-interface.md`.

For each adjacent descent $p \to r$, obstruction is a finite phase condition modulo $\prod_{\ell \le r} \ell$ and is classified as either grid failure or endpoint-puncture blocking. The current descent tree has twenty actual transition nodes, all successful, with no actual blocked node. This does not close the global theorem; it reduces any future obstruction to the SAE finite package, PDEC endpoint phase row, or ColumnCRT displacement row. The certificate skeleton builder

`docs/monograph/prime-matrix-rpz-certificate-skeleton-package.md`

materializes these rows: two endpoint SAE candidates, two endpoint PDEC/ColumnCRT rows, and three lower-descent grid-failure PDEC/ColumnCRT rows. These are certificate obligations, not completed exclusions. The endpoint SAE finite certificate

`docs/monograph/prime-matrix-rpz-endpoint-sae-finite-certificate.md`

checks that the two endpoint candidates have possible load two but actual load zero in the current BCB ledger. Hence the current finite endpoint SAE line is vacuously closed; global SAE exclusion is still not proved. The lower grid-failure certificate

`docs/monograph/prime-matrix-rpz-lower-grid-fail-avoidance-certificate.md`

reduces every adjacent descent $p \to r$ to the exact test $\delta = -(a-1)(p-r) \bmod r$ and grid success iff $\delta \le p - r$. The current descent tree has twenty actual transition nodes and none hits grid failure.

The remaining global task is to prove this phase inequality for formal descent paths, or route its failure to PDEC/ColumnCRT. The formal path note

`docs/monograph/prime-matrix-rpz-formal-descent-phase-inequality.md`

sharpens this statement: once a formal descent path exists, the phase inequality is automatic from interval containment. The true remaining task is path existence for every formal counterexample branch, or exclusion of the first obstruction phase through SAE/PDEC/ColumnCRT. The first-obstruction dichotomy

`docs/monograph/prime-matrix-rpz-first-obstruction-dichotomy.md`

removes endpoint puncture as a genuine obstruction: a contained lower row containing the endpoint ap would force $r \mid a$, contradicting the roughness condition needed for the endpoint to revive. Hence every non-descending branch has a first grid-failure seam phase, which must be discharged by SAE/PDEC/ColumnCRT certificates. The first-grid-failure seam certificate

`docs/monograph/prime-matrix-rpz-first-grid-fail-seam-certificate.md`

puts this obstruction in normal form. If $g = p - r$ and $g < \delta < r$, the bad row is the union of two adjacent lower-row caps of lengths δ and $r + g - \delta$, with conserved missing mass $r - g$ and at most the right endpoint ap as an r-rough survivor. The certificate produces twelve seam phase rows covering all 1752 grid-failure phases in the current ledger. It now also records the exact PDEC support row for each seam:

$$S_\tau = \{a \bmod Q : \ a \equiv \rho \pmod r\}, \qquad F_\tau = \mathbf{1}_{a\equiv\rho \pmod r} - 1/r,$$

whose non-zero Fourier support is contained in the frequencies jQ/r, $1 \le j < r$. The same certificate splits the endpoint ap into 1348 phases already killed by lower labels and 404 Q-unit endpoint phases. Thus the next narrow gate is no longer support extraction, but either an endpoint-PDEC upper bound $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for these same bad-window families or a ColumnCRT displacement certificate with explicit selectors Π, λ and threshold L_D. The unit-endpoint gate certificate

`docs/monograph/prime-matrix-rpz-unit-endpoint-columncrt-gate.md`

pushes this one step further. For a unit endpoint seam,

$$c \equiv ap \equiv p\rho \equiv r + (p - r) - \delta \pmod r,$$

so the endpoint lies in a fixed non-trivial lower column. If $\pi = hr + c$ is a same-column prime witness, then the endpoint lower row satisfies $H \equiv -cr^{-1} \pmod p$, and the displacement $d = h - H \pmod p$ is independent of the row phase and non-zero. In the current ledger all twelve gate rows have explicit witnesses and non-zero displacements, covering all 404 unit endpoint phases. This routes persistent unit endpoint seams to fixed non-zero ColumnCRT candidates; it still does not exclude the ColumnCRT threshold L_D or prove that formal counterexample families avoid these gates. The threshold obstruction audit

`docs/monograph/prime-matrix-rpz-columncrt-threshold-obstruction.md`

then shows why this cannot be closed by merely lowering L_D. Inside a fixed unit gate, all unit endpoint residues already have the same label p and the same non-zero displacement class $d \bmod p$, so the intrinsic load of that class is

$$R_{p,d} \ge \prod_{\ell<r}(\ell - 1).$$

In the present ledger the maximum intrinsic load is 48, and the test value $L_D = 2$ would send ten gate rows, covering 400 phases, into ColumnCRTDefect rather than exclude them. Thus the remaining legitimate closures are a formal-family avoidance theorem, an endpoint-PDEC upper bound, or an independent exclusion theorem for the resulting ColumnCRTDefect. The three-route audit

`docs/monograph/prime-matrix-rpz-three-route-closure-audit.md`

compares these options under the same ledger. It finds no closed route yet, but ranks formal-family avoidance first: the present lower-descent ledger has twenty actual transition nodes and no grid-failure node, while the endpoint-PDEC route still lacks an admissible U_{CRT} upper bound and the ColumnCRT route has been reduced to an independent defect-exclusion theorem. The formal phase automaton

`docs/monograph/prime-matrix-rpz-formal-phase-automaton.md`

turns the first route into an explicit start-phase admission problem. For each prime level p it defines an accepted set $A_p \bmod P(p)$: phases in A_p descend canonically through successive prime levels to $p = 2$, while phases outside A_p hit a first-grid-fail seam and return to the already materialized exits. In the current ledger all six BCB-Core start rows belong to their accepted sets. The global missing theorem is that every formal counterexample-family start phase lies in A_p, or else is captured by the seam exits. The rejected-phase absorption certificate

`docs/monograph/prime-matrix-rpz-rejected-phase-absorption.md`

removes the second ambiguity within the present automaton range. It checks all 27924 rejected phases, finds twelve distinct first-fail seam rows, and verifies that every one of them is already materialized by the first-grid-fail seam certificate. Hence the RPZ formal route is now a strict two-way routing statement: an accepted start phase descends to $p = 2$, while a rejected start phase enters the named seam/PDEC/ColumnCRT certificate chain. This is an absorption theorem, not an exclusion theorem; the remaining hard point is the actual discharge of those seam/PDEC/ColumnCRT exits, or a proof that formal counterexample starts always lie in A_p. The seam exit pressure ledger

`docs/monograph/prime-matrix-rpz-seam-exit-pressure-ledger.md`

then compresses this remaining exit. The 27924 rejected phases reduce to twelve seam rows; within those rows 1348 endpoint phases are already killed by lower labels, while the remaining 404 unit endpoint phases fall into ten fixed non-zero ColumnCRT displacement classes, with maximum aggregated load 96. Thus the next gate is no longer a phase search: one must either prove formal-family avoidance of the twelve seam rows, provide $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for their single-residue PDEC supports, or prove an independent exclusion theorem for the ten ColumnCRT displacement classes. The symbolic ladder certificate

`docs/monograph/prime-matrix-rpz-symbolic-ladder-certificate.md`

rewrites the preferred avoidance route as local adjacent-prime digits. For each adjacent pair $p > r$, with $g = p - r$, it uses

$$\delta_p(a) = -(a-1)g \pmod{r}, \qquad \delta_p(a) \le g$$

as the exact success digit. Failure of this inequality is precisely the first-grid-fail seam already materialized above. The certificate verifies the resulting counting formula by full enumeration through modulus $P(19) = 9699690$ and symbolically records the same ladder through $p = 97$. The remaining theorem is now sharply stated: derive these digit inequalities from the formal counterexample construction itself, not from a finite phase enumeration. The BCB start-digit ledger

`docs/monograph/prime-matrix-rpz-bcb-start-digit-ledger.md`

extracts these digits for the current BCB-Core starts. All twenty digit nodes are safe, but ten of them have zero margin $\delta_p(a) = p - r$. This is a useful negative result for the proof strategy: the desired avoidance theorem cannot be closed by a coarse positive margin estimate. It must use exact congruence information for the start row modulo each lower prime r. The accepted lower-row selector ledger

`docs/monograph/prime-matrix-rpz-bcb-accepted-row-selector.md`

weakens the required positive statement to the exact selector condition

$$C_h(J) \cap A_h \neq \varnothing, \qquad C_h(J) = \{m : \ J \supset [(m-1)h + 1, mh]\}.$$

For the current BCB-Core samples all five core intervals have such a selector, and all six candidate lower rows are accepted. This is still only a current-ledger certificate; the global theorem must prove selector existence for every formal BCB core interval, or else route the all-rejected selector failure back to the seam/PDEC/ColumnCRT exits. Finally, the selector gap ledger

`docs/monograph/prime-matrix-rpz-selector-gap-threshold.md`

splits selector existence into two verifiable mechanisms. If the number of complete candidate rows in $C_h(J)$ exceeds the maximum cyclic rejected gap of A_h, length alone forces a selector. In the current samples this explains two of the five BCB cores; the remaining three are short-candidate cases and require exact endpoint phase information. Thus the next proof obligation is not merely to enlarge J, but to control the endpoint phase of short cores relative to the rejected gaps of A_h. The short-candidate endpoint phase ledger

`docs/monograph/prime-matrix-rpz-short-candidate-phase-ledger.md`

performs this refinement at modulus $hP(h)$ rather than just modulo h. In the current ledger the three short-candidate BCB records all hit an accepted selector. However, the full (h, ℓ) phase families still contain all-rejected endpoint phases: 588 for $(h, \ell) = (7, 13)$, 11880 for $(11, 19)$, and 344760 for $(13, 25)$. These rejected phases are not unnamed escapes; each carries a first-failure seam key and therefore returns to the seam/PDEC/ColumnCRT exits. The remaining theorem-level gate is consequently exact: prove that the formal BCB construction avoids these all-rejected $hP(h)$ endpoint phases, or close the corresponding seam/PDEC/ColumnCRT certificates. The follow-up forbidden-block ledger

`docs/monograph/prime-matrix-rpz-short-phase-block-formula.md`

removes another layer of enumeration. Since the relevant short cores have length $\ell < 2h$, a starting endpoint u contains at most one complete h-row. For a lower row m, the endpoint interval that contains it is exactly

$$mh - \ell + 1 \leq u \leq (m-1)h + 1,$$

of width $\ell - h + 1$. Hence the all-rejected endpoint set is the disjoint union of such blocks over rejected row phases $m \bmod P(h)$. The formula matches enumeration in all three short families, and the current endpoints have positive distance from the rejected endpoint union. The remaining non-enumerative input is therefore narrower still: derive from the formal BCB construction that the induced candidate row phase belongs to A_h, or else close its named first-failure exit. The BCB candidate phase identity ledger

`docs/monograph/prime-matrix-rpz-bcb-candidate-phase-identity.md`

then makes the induced row phase explicit. If the top zero row is R, the top prime is P, the half-prime is h, the plateau extremes are s_-, s_+, and the tail threshold is T, the forced core is

$$L = (R-1)P + 1 + s_- + T, \qquad U = RP + s_+ - T.$$

The complete lower rows inside this core are exactly

$$m_{\min} = \left\lfloor \frac{L+h-2}{h} \right\rfloor + 1, \qquad m_{\max} = \left\lfloor \frac{U}{h} \right\rfloor .$$

Equivalently, after writing $P = Qh + d$, these phases depend only on $R \bmod hP(h)$ and the plateau parameters. In the current ledger the formula matches all five BCB samples and all six induced lower row phases lie in A_h. This is the sharp remaining input: prove the same residue implication for every formal BCB counterexample, or send any rejected induced phase to its first-failure exit. The accepted-preimage ledger

`docs/monograph/prime-matrix-rpz-bcb-accepted-preimage-ledger.md`

enumerates this final residue condition for the current BCB parameter families. The $P = 13, h = 5$ family has no bad residue. The other four families still have bad residues, split between no-candidate endpoint failures and all-rejected candidate phases. The actual top-row residues in the ledger all lie in the selector preimage, with positive distance from the bad set where the bad set is nonempty. Thus the residue preimage computation is now explicit, but the proof is still not global: the formal counterexample construction must be shown to land in the selector preimage, or the bad residue must be discharged through its named BCB endpoint or first-failure seam/PDEC/ColumnCRT exit. The core-run obstruction ledger

`docs/monograph/prime-matrix-rpz-bcb-core-run-obstruction.md`

adds a stronger upstream test for the no-TailAnchor branch. In that branch BCB-Core forces J_T itself to be an h-sieve zero interval, so its length cannot exceed the maximum cyclic run of integers divisible by primes $\leq h$ modulo $P(h)$. For the current layers these exact maxima are

$$h = 5 : 5, \qquad h = 7 : 9, \qquad h = 11 : 13, \qquad h = 13 : 21.$$

The five current BCB cores have lengths $9, 13, 15, 19, 25$, respectively, so every one of them exceeds the corresponding low-sieve run bound, with minimum margin 4. Thus the current finite BCB no-TailAnchor samples are obstructed before the accepted-preimage question is reached. The global theorem-level input is now a Jacobsthal-type low-sieve run bound for the formal layers; without such a bound, the remaining bad cases still route to the named endpoint and first-failure exits. This input is isolated in

`docs/monograph/prime-matrix-rpz-bcb-jacobsthal-closure-interface.md`.

Writing $G(h)$ for the maximum length of a consecutive interval all of whose points are divisible by some prime $\leq h$, the exact BCB closure condition is

$$G(h) < P + m - 1 - 2T,$$

where m is the sliding platform length. Under this inequality the no-TailAnchor BCB branch closes immediately; if it fails, the proof must keep the TailAnchor, endpoint, first-failure, PDEC, and ColumnCRT exits active. The risk scan

`docs/monograph/prime-matrix-rpz-bcb-jacobsthal-risk-scan.md`

checks this linear closure route against the Ziller–Morack primorial Jacobsthal data. It shows that the current finite samples are safe, but the naive global specialization $m = 5$, $T = 4$, and only $P > 2h$ fails already at $h = 43$: there $G(h) = 89$, the minimal top prime is $P = 89$, and the core length is only 85. Thus the finite core-run obstruction cannot be promoted to a global theorem by a blanket linear bound. A global proof must either force longer platforms m, prove stronger restrictions on the formal BCB parameter set, or close the TailAnchor/endpoint/first-failure exits. The platform-threshold ledger

`docs/monograph/prime-matrix-rpz-bcb-platform-length-threshold.md`

rewrites this requirement as the exact integer condition

$$m \geq G(h) - P + 2 + 2T.$$

For $T = 4$, the first substantive unsafe layer with $h \geq 5$ is again $h = 43$, where the required platform length is 10 rather than 5. The next attack must therefore prove platform-length growth or prove that short platforms trigger one of the named exits. The short-platform embedding route

`docs/monograph/prime-matrix-rpz-bcb-short-platform-embedding-route.md`

handles the second alternative. If $N = P + m - 1 - 2T \leq G(h)$ and no TailAnchor is present, then the forced core J_T is an h-sieve zero interval of length N and hence must be embedded inside an extremal low-sieve covered block modulo $P(h)$. Therefore its left endpoint lies in a finite Jacobsthal endpoint set $E_{h,N}$. Persistent occurrence of such an endpoint phase is no longer an unnamed escape: it is routed to SAE if sparse, or to PDEC/ColumnCRT if persistent. The accompanying certificate

`docs/monograph/prime-matrix-rpz-bcb-short-platform-embedding-certificate.md`

materializes the max-block subfamily $E_{h,N}^{\max}$ by reconstructing CRT left endpoints from the Ziller–Morack modulus representation. For the current $m = 5, T = 4$ interface, the first non-trivial short platform is $h = 43$, $P = 89$, $N = 85$, with 240 distinct max-block endpoint phases; across the scanned table, 1197 reconstructed longest blocks pass the coverage and boundary checks. This remains a certificate for the longest-block subfamily, not a proof that the full endpoint set $E_{h,N}$ has been exhausted.

These obligations are not typographical tasks. They are the remaining theorem-level gates. Until they are closed, the monograph may be submitted only as a synthesis, dependency, and verification manuscript, not as a final unconditional proof of the prime-matrix theorem, prime-pair assertions, or RH.

Chapter 6

Referee Closure Audit

This chapter records the strict closure standard used for the whole monograph. A contradiction-field chain is considered closed only if its entrance, decomposition, exits, and global contradiction are all verified under the same load convention.

Definition 6.1 (Closure certificate)**.** A closure certificate for a contradiction field consists of four data:

1. an entrance theorem sending every counterexample into the field;
2. a disjoint or source-deleting decomposition into named exits;
3. for each exit, an absorption, descent, transfer, external theorem, or finite certificate;
4. a global comparison showing that final load is simultaneously bounded below and above in incompatible ways.

If any item is replaced by an unverified structural input, the conclusion is conditional on that input.

Theorem 6.2 (Current monograph closure)**.** *The present manuscript closes the audit and dependency structure of the two contradiction-field programs: all known entrances, decompositions, exits, external inputs, finite certificates, and referee obligations are explicitly named. It does not close the final unconditional theorem status of either program.*

Proof. For the prime-matrix program, the A/B appendix gives the entrance reduction from a row or column counterexample to a Structured-EHPD bad configuration. Inside the BPN branch, the low-hole bucket submodule is now closed by the low-range certificate, the narrow-band collision certificate, the two finite tail certificates, and the Rosser–Schoenfeld explicit tail package recorded above. This is a genuine submodule closure. The D-structure exclusion and the remaining PDEC/SAE/Rankin global exits remain decisive inputs requiring independent line-by-line review. Therefore the whole chain is closed as a reduction and certificate package but not as a final theorem.

For the RH program, the RH manuscript has completed U1–U5: statement wording, AEX acceptance, EXT localization, manuscript wording, BibTeX/PDF compilation, and warning-free LaTeX logs. The final audit still requires independent referee verification of every local theorem and controlled exit. Therefore the chain is closed as a verification package but not as a final unconditional RH theorem.

The status table and the referee audit in `docs/monograph/` record exactly which pieces are proved, reduced, external, computational, or awaiting review. Thus the monograph's logical boundary is explicit and closed, while promotion to final theorem status remains conditional on external verification. □

Theorem 6.3 (Internal line-by-line audit)**.** *The author-side line-by-line audit has found no new item classified as a mathematical block. The remaining blocks are referee blocks: Prime Matrix final promotion requires independent acceptance of the D-structure/Tail-log4/finite-verification interface, and RH final promotion requires independent acceptance of all controlled exits in the RH contradiction-field manuscript.*

Proof. The internal audit matrix assigns one of four labels to each link. No link is a math-block.

The terminal entries PM-16 and RH-16 remain referee-blocks. An author-side check cannot replace independent referee verification. Therefore the internal audit is complete, but final unconditional theorem promotion is not. □

Remark 6.4 (No overclaim principle)**.** The monograph may state the shared contradiction-field method and the completed audit closures. It must not state unconditional proofs of RH or of the prime-matrix row/column theorem until the remaining structured inputs have independent acceptance.

Chapter 7

Current Status and Honest Boundary

The purpose of this monograph draft is to make the shared structure visible and auditable. It is not yet a final unconditional theorem manuscript. The precise statuses are maintained in `docs/monograph/claim-status-table.md`. Any future promotion to final theorem status requires independent verification of every reduced structured input and every controlled exit.